

Topics in Statistical Inference

Ryan Martin¹

North Carolina State University

`www4.stat.ncsu.edu/~rmartin`

42nd Finnish Summer School on Stochastics & Statistics
May 27–31 2024

¹Research partially supported by the NSF, SES-2051225

- This course is about *statistical inference*
 - logic & foundations
 - theory
 - methods
- Aim is to cover a broad range of topics, with varying degrees of depth throughout
- Lots of material, so it'll be intense
- I also hope it'll be fun and we'll learn some things

- The lectures follow the material in a new book I'm writing
- It's more than 50% complete, still a few chapters to go
- What's completed is readable

Fundamentals of Statistical Inference
Intuition, Logic, Theory, and their Use

Ryan Martin
North Carolina State University
www4.stat.ncsu.edu/~rmartin

DRAFT: May 22, 2024

- 1 Background & “making sense of figures”
- 2 Information & likelihood-based methods
- 3 Bayesian methods
- 4 Decision theory
- 5 Some extras
 - reliable imprecise-probabilistic inference
 - Bayesian analysis without a model



- Dr. Ryan Martin
- PhD, 2009, Statistics, Purdue
- NCSU Statistics since 2016
- co-author of *Inferential Models*
- recent research efforts:
 - foundations of statistics
 - imprecise probability
 -

- Probability background
 - concepts and notation
 - simple examples
 - sampling distributions
 - approximations: asymptotic & numerical
- Logic of statistical inference
 - problem setup
 - deductive vs inductive logic
 - specialization to statistics
 - connection to sampling distributions

- I'll assume familiarity with probability theory
 - probability space $(\Omega, \mathcal{A}, P)$
 - Kolmogorov's axioms for P
 - direct consequences, e.g.,
 - complementation rule, $P(A^c) = 1 - P(A)$
 - monotonicity, $A \subseteq B \implies P(A) \leq P(B)$
- Interpretation? I say probability is always subjective...
- Conditional probability corresponds to updating beliefs based on additional information
- i.e., if I know that event B has occurred, then I update my beliefs about A are changed from $P(A)$ to

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

- Random variables:
 - mathematically, it's a function $X : \Omega \rightarrow \mathbb{X}$
 - practically, it represents a feature of the experimental unit
- Mathematical representation of the variables that we measure when we collect data
- Distribution of random variable X
 - probabilities about X inherited from P on Ω
 - $P_X(E) = P(X \in E) = P(\{\omega : X(\omega) \in E\})$
- Advantage is that we can directly write down the distribution of X , without considering original $(\Omega, \mathcal{A}, P)$

- Various ways to describe the distribution of X
 - cumulative distribution function (CDF)
 - corresponding mass (PMF) or density (PDF) function
 - moment-generating function (MGF)
 - ...
- Probability calculations:

$$P(X \in E) = \begin{cases} \sum_{x \in E} p_X(x) & \text{if } X \text{ is discrete} \\ \int_E p_X(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

- Expected values:

$$E\{g(X)\} = \begin{cases} \sum_x g(x) p_X(x) & \text{if } X \text{ is discrete} \\ \int g(x) p_X(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

- Any CDF defines a distribution for X
- But there are some particular kinds of probability models that are commonly used in applications
- Common models:
 - binomial
 - Poisson
 - uniform
 - gamma
 - normal
 - ...
- Details on these models will come later in examples
- Depend on parameters, generically referred to as θ

- We often have more than one variable
- Collect these as $X^n = (X_1, \dots, X_n)$
- Then we'd like to describe their *joint distribution*
- We'll focus here on *independent* random variables, so that their joint distribution is determined by the marginals
- For independent X_i 's the joint PMF/PDF is

$$p_{X^n}(x^n) = \prod_{i=1}^n p_{X_i}(x_i)$$

- Most of our examples assume random variables are *iid*, i.e., independent and identically distributed

Conditional distributions

- If we learn the value of some feature of X^n , then what's the updated distribution of some other features?
- A common case is when there's a joint PDF/PMF and interest is in conditional distribution of, say, X_2 given $X_1 = x_1$:

$$p_{X_2|X_1}(x_2 | x_1) = \frac{p_{X_1, X_2}(x_1, x_2)}{p_{X_1}(x_1)}$$

- Other cases are more complicated...²
- e.g., conditional “density” of X^n given $T(X^n) = t$ is

$$p_{X^n|T(X^n)}(x^n | t) = \frac{p_{X^n}(x^n) 1\{T(x^n) = t\}}{p_{T(X^n)}(t)}$$

²These are measure-theoretic complications

Ingredients of a stat inference problem

- We'll soon be very abstract and succinct in our setup
- But it's good to have a clear picture of what's involved
- Three main ingredients:
 - scientific objective
 - data
 - statistical model
- Probably not an exhaustive list, but this at least roughly captures the key aspects
- I'll talk briefly about each in turn...

- The scientific (or business, ...) objective provides context
- We aren't just doing data analysis in a vacuum
- This is what gives the statistician *skin in the game*
 - makes it more than a math problem
 - gives us something to lose if we're wrong
 - motivates us to question assumptions, push ourselves for the best possible solution, etc.
- Very important, but often overlooked in courses and textbooks on statistical theory
- Probably because skin in the game can't be manufactured

- The data is what we actually work with — there can be no statistical inference without data
- Mathematically:
 - hypothetical/observable data denoted by X^n
 - observed data denoted by x^n
- Critical that data be “relevant” to the objective
- How to collect relevant data is beyond our scope
- Remarks:
 - data is everywhere nowadays
 - driving advances in data science, etc
 - data size can't make up for low quality
 - “garbage in, garbage out”

- Roughly, the statistical model helps us to mathematize the scientific objectives
- Scientific questions concern the “world,” which is quantified by the posited statistical model
- Relevant features of the world correspond to features of the model parameter θ — defines our “inferential target”
- Model choice is extremely difficult, beyond our scope
- But I'd be happy to discuss this in Q&A/breaks
 - modeling considerations
 - what can be done without models
 - ...

Statistical problem statement.

$$X^n = (X_1, \dots, X_n) \stackrel{iid}{\sim} P_\theta \text{ with } \theta \text{ unknown}$$

- iid isn't that important, mostly for simplicity
- Goal: “learn” about unknown θ from observed $X^n = x^n$
- More details later, but it starts with a summary of the data
- A statistic $T_n = T(X^n)$ is a function of X^n
 - sum or average
 - order statistic or median
 - ...
- The *sampling distribution* of statistic T_n is the distribution derived from the joint distribution of X^n

Sampling distributions, cont.

Replication	Data	Statistic
1	$X_1^{(1)}, \dots, X_n^{(1)}$	$T_n^{(1)} = T(X_1^{(1)}, \dots, X_n^{(1)})$
2	$X_1^{(2)}, \dots, X_n^{(2)}$	$T_n^{(2)} = T(X_1^{(2)}, \dots, X_n^{(2)})$
$\vdots$		
r	$X_1^{(r)}, \dots, X_n^{(r)}$	$T_n^{(r)} = T(X_1^{(r)}, \dots, X_n^{(r)})$
$\vdots$		

- Schematic of a sampling distribution
- A histogram of the right-most column provides a visualization of the sampling distribution of T_n

- In principle, the sampling distribution of T_n can be derived from the joint distribution of X^n
- Can be difficult to do in general...
- One technique that works in some cases is MGFs
- Suppose $T_n = \sum_{i=1}^n X_i$, sum of independent RVs
- Then the MGF of T_n is

$$M_{T_n}(u) = E(e^{uT_n}) = \prod_{i=1}^n E(e^{uX_i}) = \prod_{i=1}^n M_{X_i}(u)$$

- If the MGF of T_n has a familiar form, then we know the sampling distribution of T_n

Exercise.

Let $X_i \stackrel{\text{ind}}{\sim} \text{Bin}(k_i, \theta)$, for $i = 1, \dots, n$, and set $T_n = \sum_{i=1}^n X_i$. Use MGFs to show that $T_n \sim \text{Bin}(m, \theta)$, where $m = \sum_{i=1}^n k_i$.

Exercise.

Let $X_i \stackrel{\text{ind}}{\sim} \text{Pois}(\theta_i)$, for $i = 1, \dots, n$, and set $T_n = \sum_{i=1}^n X_i$. Use MGFs to show that $T_n \sim \text{Pois}(\lambda)$, where $\lambda = \sum_{i=1}^n \theta_i$.

- Aside from certain special cases, the exact sampling distribution of a statistic isn't available
- Need ways to approximate the sampling distribution:
 - asymptotic, i.e., limits as $n \rightarrow \infty$
 - numerical, i.e., via computer simulations
- Both are useful, and we'll use both in this course
- Former requires notions of stochastic convergence:
 - in probability
 - in distribution
 - ...
- These have different interpretations

Definition.

$T_n \rightarrow T_\infty$ in probability as $n \rightarrow \infty$ if, for all $\varepsilon > 0$,

$$P\{|T_n - T_\infty| > \varepsilon\} \rightarrow 0, \quad \text{as } n \rightarrow \infty$$

Definition.

$T_n \rightarrow T_\infty$ in distribution as $n \rightarrow \infty$ if

$$P(T_n \leq t) \rightarrow P(T_\infty \leq t), \quad \text{as } n \rightarrow \infty$$

for all t at which the right-hand side is continuous

- Former: T_n and T_∞ are close with high probability
- Latter: sampling dist'n of $T_n \approx$ sampling dist'n of T_∞
- Former is most meaningful when T_∞ is constant and, in that case, the two are equivalent

- A particularly nice and commonly used statistic is the sample mean, $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$
- Averaging operation is stabilizing
- Below are two classical results about $\bar{X}_n$

Law of large numbers.

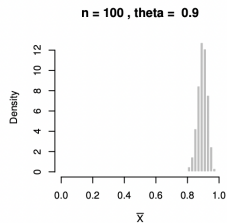
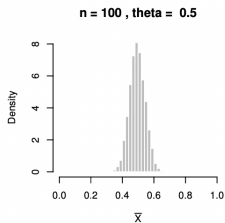
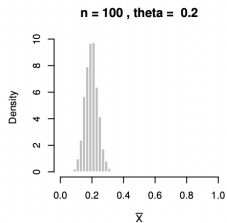
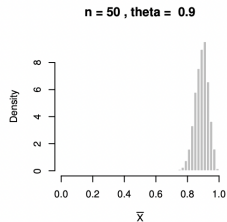
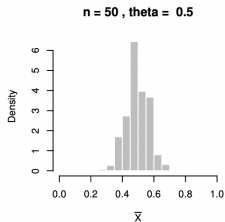
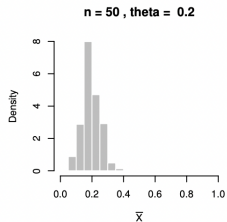
If $X_1, X_2, \dots$ are iid with mean μ , then $\bar{X}_n \rightarrow \mu$ in probability

Central limit theorem.

If $X_1, X_2, \dots$ are iid with mean μ and variance σ^2 , then

$$n^{1/2}(\bar{X}_n - \mu) \rightarrow N(0, \sigma^2) \quad \text{in distribution}$$

Sampling distributions, cont.



Exercise.

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$ and set $T_n = \max(X_1, \dots, X_n)$. Show that $T_n \rightarrow \theta$ in probability.

Exercise.

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$ and set $T_n = \max(X_1, \dots, X_n)$. Show that $n(\theta - T_n) \rightarrow \text{Exp}(\lambda)$ in distribution, and find λ .

- New convergence theorems from old?
- Intuitively, applying a nice transformation to a convergent sequence shouldn't ruin convergence
- The continuous mapping theorem makes this precise

Continuous mapping theorem.

Let T_n be a sequence converging in probability (or in distribution) to T_∞ . If g is a continuous function, then $g(T_n) \rightarrow g(T_\infty)$ in probability (or in distribution)

- Shortcoming of the in-distribution version of CMT is that it doesn't identify the limit distribution
- We can get this with a little more structure/smoothness
- The delta theorem makes this precise

Delta theorem.

Suppose that T_n satisfies $n^{1/2}(T_n - \theta) \rightarrow N(0, v(\theta))$ in distribution. If g is differentiable, and $\dot{g}(\theta) \neq 0$, then

$$n^{1/2}\{g(T_n) - g(\theta)\} \rightarrow N(0, \dot{g}(\theta)^2 v(\theta)), \quad \text{in distribution}$$

Exercise.

Let $X_1, X_2, \dots \stackrel{\text{iid}}{\sim} \text{Pois}(\theta)$, and $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$.

- 1 Use the delta theorem to find the limiting distribution of $g(\bar{X}_n)$ where $g(x) = (x - 7)^2$
- 2 There's one special instance of the situation in Part 1 above in which delta theorem can't be applied. What is that instance and what's the issue?

Exercise.

Under the delta theorem setup, suppose that $\dot{g}(\theta) = 0$. Formulate second-order delta theorem for this case.

Sampling distributions, cont.

- Sometimes we have a mix of cases involving in-probability and in-distribution convergence
- Slutsky's theorem confirms what you'd expect to happen
- Similar to the continuous mapping theorem
 - weaker assumptions
 - only applies to certain simple functions

Slutsky's theorem.

If $X_n \rightarrow X$ in distribution and $Y_n \rightarrow y$ in probability, then

$$X_n + Y_n \rightarrow X + y, \quad X_n Y_n \rightarrow X y, \quad X_n / Y_n \rightarrow X / y,$$

all in distribution; latter requires that $y \neq 0$

Exercise.

Let $X_1, X_2, \dots$ be iid with mean μ and variance σ^2 . Define the sample variance based on X^n ,

$$V_n = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2,$$

and show that $V_n \rightarrow \sigma^2$ in probability

Exercise.

Let X_n be a random variable that converges in distribution to X . Also let A_n be an event with $P(A_n) \rightarrow 1$. Show that, for any random variable Z_n ,

$$X_n 1(A_n) + Z_n 1(A_n^c) \rightarrow X \quad \text{in distribution}$$

- The following simple result is important
- Lots of applications:
 - useful for Monte Carlo methods
 - drives much of what we do with inference

Probability integral transform.

Let F be a continuous, strictly increasing CDF.

- If X has CDF F , then $F(X) \sim \text{Unif}(0, 1)$
- If $U \sim \text{Unif}(0, 1)$, then $F^{-1}(U)$ has CDF F

Proof.

Exercise.



- The above is an example of a *pivot* or *pivotal quantity*
- If data X^n is iid P_θ , then a *pivot* is a function $Q_\theta(X^n)$ whose sampling dist'n doesn't depend on θ
- General rules for constructing pivots are scarce
- Easy to do in structured models, e.g., location–scale
- A function $Q_\theta(X^n)$ is an *asymptotic pivot* if its limit dist'n doesn't depend on θ
- e.g., standardizing a normal limit distribution

Exercise.

Let $X_1, \dots, X_n$ be iid $\text{Exp}(\theta)$.

- 1** Find $a_n(\theta)$ and $b_n(\theta)$ such that

$$Q_\theta(X^n) = \frac{\sum_{i=1}^n X_i - a_n(\theta)}{b_n(\theta)}$$

is an approximate pivot, i.e., has non-degenerate limit distribution that doesn't depend on θ .

- 2** Same as above, but now for

$$Q_\theta(X^n) = \frac{\max_{i=1, \dots, n} X_i - a_n(\theta)}{b_n(\theta)}$$

- Numerical approximations are available too
- Based on *Monte Carlo* methods
- Effectively just an implementation of my schematic
- To carry this out, however, we need to know the distribution of X^n , including the parameters
 - in theory this often isn't a problem
 - but makes computation more expensive
 - more details later...
- Bootstrap is another useful tool

Questions, etc?

“Making sense of figures”

For everyone who does habitually attempt the difficult task of making sense of figures is, in fact, assaying a logical process of the kind we call induction, in that he is attempting to draw inferences from the particular to the general, from the sample to the population. —Fisher

- Setup:³
 - “ $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P_\theta$ with unknown $\theta \in \mathbb{T}$ ”
 - goal is to “learn” about θ from observations $X^n = x^n$
- Unlike in mathematics, we can’t *deduce*, solely from the given info, statements about θ that are surely true
- The point is we can’t control the data
- The data might not be perfectly representative of the world
 - we might do everything right
 - but we reach a wrong conclusion
- So, the familiar deductive logic doesn’t work here
- Fortunately, there are other logical systems

³Nontrivial assumption: the model is “correct” ...

- *Inductive logic* is what we need for statistical inference
- We do this all the time without second thought
- Example:
 - I claim a coin is fair
 - to test my claim, you toss the coin 10 times
 - if you observe 10 heads, then you doubt my claim
- What's your reasoning?
 - 10-out-of-10 heads is improbable if my claim were true
 - you don't expect improbable events to occur
 - so your conclusion is that my claim is wrong
- This is intuitive, but there's a lot going on...

- There are a number of requirements
 - relevant summary of the data
 - means to evaluate “probability”
 - a rule for classifying events as “improbable”
- Straightforward in the coin flipping example
- More effort needed to flesh this out in general
- But a sampling distribution connection should be clear
- Key points:
 - can't guarantee conclusions are true
 - force of the argument is it's inherent reliability
 - i.e., conclusions are false iff an “improbable” event occurs which, by definition,⁴ is improbable

⁴This assumes a sound model...

- Again: inductive logic doesn't guarantee true conclusions
- e.g., not impossible to get 10-out-of-10 heads from a fair coin
- Some might say this makes statistical inference “soft”
- But (Fisher and) I disagree:
 - the real world is dirty
 - deduction stays clean/pure by working in a fantasy world
 - induction's strength is that it can handle the dirt

In deductive reasoning all knowledge attainable is already latent in the postulates... In inductive reasoning we are performing part of the process by which new knowledge is created. —Fisher

- General inductive logic for statistical inference?
- To the relevant questions (about θ), we want to assign a data-driven answer, say $\text{ans}(X^n)$
- Answer could be wrong or not-wrong, but since we don't know θ we can't make this determination
- Ideally,⁵ we find procedures $\text{ans}(\cdot)$ such that

$$P_{\theta}\{\text{ans}(X^n) \text{ is "wrong"}\} \text{ is small for all } \theta$$

- Definition of “wrong” must be non-data-dependent
- The above is a sampling distribution property

⁵Typically, we can't achieve this ideal, so some relaxation is needed...

- Suppose the goal is to estimate $\phi(\theta)$
- An *estimator* is a function $\hat{\phi}_n = \hat{\phi}(X^n)$ of the data X^n
- i.e., $\hat{\phi}_n$ an “answer” to the question “what is $\phi(\theta)$?”
- The answer is “wrong” if $|\hat{\phi}_n - \phi(\theta)| > \varepsilon$, where $\varepsilon > 0$ is pre-specified by the user
- Then we’d want $\hat{\phi}$ to be such that

$$P_{\theta}\{\underbrace{|\hat{\phi}(X^n) - \phi(\theta)|}_{\text{“answer is wrong”}} > \varepsilon\} \quad \text{is small for all } \theta$$

- Scientific objective/skin in the game motivates us to find $\hat{\phi}$ to optimize all aspects

- Might be challenging to do so at a fixed n /all θ
- Relaxation: probability is vanishingly small as $n \rightarrow \infty$

Definition.

$\hat{\phi}_n = \hat{\phi}(X^n)$ is consistent if

$$P_{\theta}\{|\hat{\phi}(X^n) - \phi(\theta)| > \varepsilon\} \rightarrow 0 \quad \text{for all } \varepsilon > 0 \text{ and all } \theta$$

Uniformly consistent if above convergence is uniform in θ

Definition.

$\hat{\phi}_n = \hat{\phi}(X^n)$ is r_n -consistent, for $r_n \rightarrow 0$, if

$$P_{\theta}\{|\hat{\phi}(X^n) - \phi(\theta)| > M_n r_n\} \rightarrow 0 \quad \text{for all } M_n \rightarrow \infty \text{ and all } \theta$$

Uniformly r_n -consistent if above convergence is uniform in θ

Exercise.

Under the conditions of the central limit theorem, show that the sample mean is a $n^{-1/2}$ -consistent estimator of the expectation

Exercise.

Let $X_1, \dots, X_n$ be iid $\text{Unif}(0, \theta)$. Show that the sample max is a n^{-1} -consistent estimator of θ

- Lots of ways to construct (good/decent) estimators
 - sample moments $\rightsquigarrow$ method of moments
 - sample quantiles
- I'll skip here, but details in Ch. 2 of book
- Later we'll cover “optimal” estimators
- It's also easy to see how forcing “probability of a wrong answer” to be small fits with other statistical goals:
 - hypothesis tests
 - confidence sets
- I'll focus here on something bigger-picture which immediately implies these familiar notions/results

Uncertainty quantification

- Assertion / hypothesis / proposition " $\phi(\theta) \in B$ "
- Motivated by the scientific problem
- Binary yes/no answer to this question: $1(\hat{\phi}_n \in B)$
- Not fully satisfactory, no uncertainty quantification
- We can and should do better...

Statisticians want numerical measures of the degree to which data support hypotheses. —Hacking

[Fisher] was the world's master of quantifying uncertainty, having developed many of the... procedures for doing so. —Pearl

- Goal: assign data-dependent numerical values to hypotheses
- Two distinct (but related) interpretations:
 - measures *support for* the hypothesis
 - measures *plausibility of* the hypothesis ✓
- $T_{\vartheta}(X^n)$ is a function that measures the (in)compatibility of data X^n with a generic value ϑ of θ
- Large values of $T_{\vartheta}(X^n)$ mean X^n and ϑ are incompatible
- e.g., $T_{\vartheta}(X^n) = h(|\hat{\theta}(X^n) - \vartheta|)$, for h non-decreasing

Definition.

A p-value function for θ , given $X^n = x^n$, is

$$\pi_{x^n}(\vartheta) = P_{\vartheta}\{T_{\vartheta}(X^n) \geq T_{\vartheta}(x^n)\}, \quad \vartheta \in \mathbb{T}$$

- Remarks:

- $\pi_{x^n}(\vartheta) \neq$ probability that “ $\theta = \vartheta$ ” given $X^n = x^n$!
- $\pi_{x^n}(\vartheta) =$ plausibility of “ $\theta = \vartheta$ ” given $X^n = x^n$!
- probability is two-sided, plausibility is one-sided
- this can be flipped into a “measure of support”
- there are other names for π in the literature

- Things I haven't explained yet:

- where $T_{\vartheta}(X^n)$ comes from
- why the values returned by the p-value are meaningful
- ...

- The former comes a bit later, the latter comes shortly

P-value function, cont.

- Example...

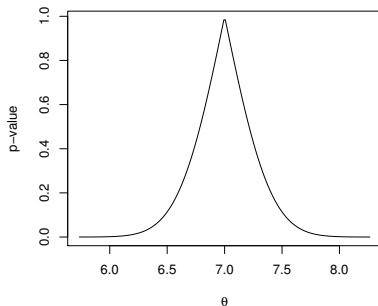
- $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$

- $T_{\vartheta}(X^n) = |\bar{X} - \vartheta|$

- P-value function

$$\vartheta \mapsto 2\{1 - \Phi(n^{1/2}|\bar{x} - \vartheta|)\}$$

- Plot: $n = 10$, $\bar{x} = 7$

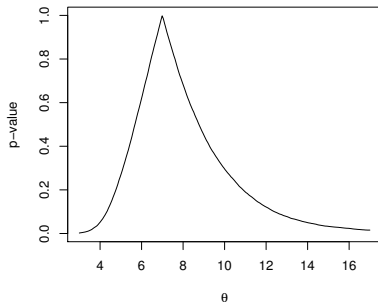


P-value function, cont.

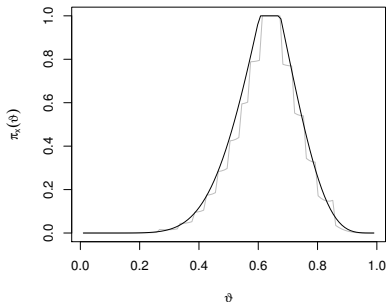
- Example...
- $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\theta)$
- $T_{\vartheta}(X^n) = |\bar{X} - \vartheta|$
- P-value function

$\vartheta \mapsto$ exercise
no formula but...

- Plot: $n = 10$, $\bar{x} = 7$



- Example...
- $X \sim \text{Bin}(n, \theta)$
- $T_{\vartheta}(X) = \dots$
- P-value function
 - Clopper–Pearson
 - likelihood ratio
- Plot: $n = 14, x = 9$



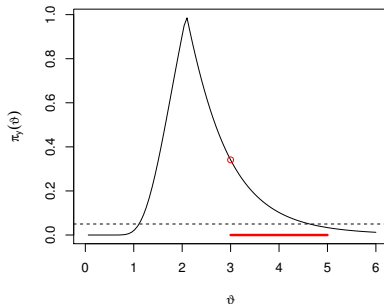
- Extend the p-value function to general hypotheses?
- Let $H \subseteq \mathbb{T}$ denote a generic hypothesis about θ
- Define extended p-value function

$$\pi_{x^n}(H) = \sup_{\vartheta \in H} \pi_{x^n}(\vartheta), \quad H \subseteq \mathbb{T}$$

- If $H = \{\vartheta\}$, then we recover the original p-value
- The plausibility of “ $\theta \in H$ ” determined by the most-plausible parameter value in H
- Support measure: $H \mapsto 1 - \pi_{x^n}(H^c)$

P-value function, cont.

- Plot $\vartheta \mapsto \pi_{X^n}(\vartheta)$
- Hypothesis $H = [3, 5]$
- $\pi_{X^n}(H)$ via optimization
- H is assessed by its “most plausible” value



- To-be-justified proposal:
 - $\pi_{X^n}(H)$ measures plausibility of H
 - if $\pi_{X^n}(H)$ is small, then inclined to doubt H
- Since plausibility is one-sided, the “answer is wrong” iff H is true and $\pi_{X^n}(H)$ is small
- Does the above event have small probability? Yes!
- Proof uses the *probability integral transform*

Theorem.

$$\sup_{\theta \in H} P_{\theta} \left\{ \underbrace{\pi_{X^n}(H) \leq \alpha}_{\text{“answer is wrong”}} \right\} \leq \alpha, \quad \alpha \in [0, 1]$$

Corollary.

- 1** The test that rejects hypothesis H iff $\pi_{X^n}(H) \leq \alpha$ controls the Type I error probability at level α , i.e.,

$$\sup_{\theta \in H} P_{\theta} \{ \pi_{X^n}(H) \leq \alpha \} \leq \alpha$$

- 2** The set

$$C_{\alpha}(x^n) = \{ \vartheta : \pi_{x^n}(\vartheta) > \alpha \}$$

is a $100(1 - \alpha)\%$ confidence set, i.e.,

$$\sup_{\theta} P_{\theta} \{ C_{\alpha}(X^n) \not\ni \theta \} \leq \alpha$$

- I've not been specific (yet) about the p-value construction
- i.e., I didn't tell you what T could/should be
- The reason is that this choice doesn't affect the basic properties above — they hold for all such T
- But the choice of T does matter
 - we don't know what it means to be “good”
 - it would look out-of-the-blue at this point
- Roughly, the “best” choice of T will
 - make efficient use of the data
 - and have the most tightly concentrated p-value

- It's rare that p-values have closed-form expressions
- Usually we'll need numerical methods/approximations
- Simple/naive Monte Carlo method

$$\pi_{x^n}(\vartheta) \approx \frac{1}{M} \sum_{m=1}^n 1\{T_{\vartheta}(X_{|\vartheta,m}^n) \geq T_{\vartheta}(x^n)\}, \quad \vartheta \in \mathbb{T},$$

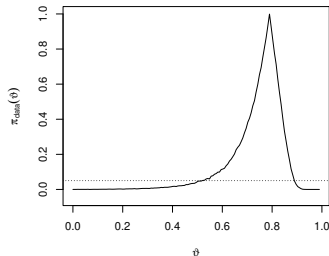
where $X_{1|\vartheta,m}, \dots, X_{n|\vartheta,m} \stackrel{\text{iid}}{\sim} P_{\vartheta}$, for $m = 1, \dots, M$

- If T defines a pivot, not uncommon, then we can use the same set of samples for all ϑ values

P-value function, cont.

Example.

Bivariate normal data with means and variances 0 and 1, resp, but with unknown correlation $\theta \in [-1, 1]$. Efron (1982) considers (standardized) data consisting of students' undergrad GPA and LSAT scores. Sample correlation is 0.776. Plot shows the Monte Carlo-based p-value function for θ . Clear, e.g., that GPA & LSAT aren't independent...



- Statistics is different from mathematics
- Inductive vs deductive logic
 - latter is “stronger”
 - but former is focused creating new knowledge
- Sampling distribution properties are key to carrying out this inductive reasoning
- My broad proposal is to use p-values for inference...
- You're likely aware of the stigma surrounding p-values
 - some say that p-values are bad for various reasons
 - this is largely due to misunderstanding, expecting p-values to do something that they're not designed to do
 - a consequence of statisticians not being clear

Exercise.

Instead of θ itself, suppose interest is in $\phi = \phi(\theta)$. Define a *marginal p-value function* for ϕ as

$$\pi_{X^n}^{\phi}(\varphi) = \sup_{\vartheta: \phi(\vartheta) = \varphi} \pi_{X^n}(\vartheta), \quad \varphi \in \phi(\mathbb{T})$$

Show that

$$C_{\alpha}^{\phi}(X^n) = \{\varphi : \pi_{X^n}^{\phi}(\varphi) > \alpha\}$$

is a $100(1 - \alpha)\%$ confidence set for $\phi(\theta)$

Exercise.

Show that the extended p-value function is sub-additive, i.e.,

$$\pi_{X^n}(H_1 \cup H_2) \leq \pi_{X^n}(H_1) + \pi_{X^n}(H_2), \quad H_1 \cap H_2 = \emptyset$$

Questions, etc?

- Information and data reduction
 - sufficiency & minimal sufficiency
 - ancillary & complete statistics
 - Fisher information
 - Cramer–Rao bound
 - connections to sufficiency
 - conditioning, etc
- Likelihood-based methods
 - likelihood function
 - maximum likelihood estimators
 - relative likelihood & p-values
 - hypothesis tests & confidence sets
 - asymptotic properties

- How to summarize data without loss of information?
- Classical results (mostly due to Fisher):
 - my coverage of this will be brief
 - but can't be skipped — it's important
- Keep in mind that reduction without loss of information is only with respect to a given model
- The things we throw away here (e.g., residuals) are needed to assess the quality of the model

- Recall, X^n is an iid sample from P_θ , θ unknown
- Roughly, a statistic $T = T(X^n)$ is sufficient if, given its value, we could generate a new data set having the same info as X^n
- The obstacle to generating a new data set similar to X^n is that we don't know θ
- So, somehow we need θ to drop out...

Definition.

A statistic T is *sufficient* for the model $\{P_\theta : \theta \in \mathbb{T}\}$ if the conditional distribution of X^n , given $T = t$, doesn't depend on θ

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$, then $\sum_{i=1}^n X_i$ is sufficient

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Pois}(\theta)$, then $\sum_{i=1}^n X_i$ is sufficient

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$, then $\sum_{i=1}^n X_i$ is sufficient

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$, then $\max X_i$ is sufficient

- The definition of sufficiency isn't so convenient to work with, i.e., it requires guess-and-check
- This would be difficult in multiparameter cases
- Can we directly find a sufficient statistic?

Factorization theorem.

Let $p_\theta(x^n)$ denote the joint PDF/PMF of X^n , given θ . A statistic T is sufficient iff there exists functions h and g_θ such that

$$p_\theta(x^n) = h(x^n) g_\theta\{T(x^n)\} \quad \text{all } \theta \text{ and all } x^n$$

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, then $(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2)$ is sufficient

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Gam}(\alpha, \beta)$, then $(\prod_{i=1}^n X_i, \sum_{i=1}^n X_i)$ is sufficient

Exercise.

More generally, let $X_1, \dots, X_n$ be iid from a K -dimensional *exponential family* model with PDF/PMF

$$p_{\theta}(x) = h(x) \exp\left\{ \underbrace{\eta(\theta)^{\top} t(x)}_{\sum_{k=1}^K \eta_k(\theta) t_k(x)} - B(\theta) \right\}$$

Show that $\sum_{i=1}^n t(X_i)$ is sufficient

Exercise.

Show that the following are exponential families:

- 1 $\text{Bin}(n, \theta)$ with n known
- 2 $\text{Pois}(\theta)$
- 3 $N(\mu, \sigma^2)$ with $\theta = (\mu, \sigma^2)$
- 4 $\text{Gam}(\alpha, \beta)$ with $\theta = (\alpha, \beta)$

Exercise.

Show that the following are not exponential families:

- 1 $\text{Cauchy}(\theta, 1)$
- 2 $\text{Unif}(0, \theta)$

- There are lots of sufficient statistics
- Lower dimensional is better, and “best” is if dimension of T matches dimension of θ
- But it's not obvious how far dimension reduction can go

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, \theta^2)$, then $(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2)$ is the sufficient statistic suggested by the factorization theorem

Minimal sufficient statistics

- A *minimal sufficient statistic* is one that's a function of every other sufficient statistic
- Not such a useful definition
- There's a characterization available, but I'll skip it
- Relevant take-away points:
 - for regular exponential families, $\dim(T_{\text{minsuff}}) = \dim(\theta)$
 - for basically every other case, $\dim(T_{\text{minsuff}}) \gg \dim(\theta)$

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Cauchy}(\theta, 1)$, then the order statistics are minimal sufficient, i.e., basically no reduction is possible

- Opposite to sufficiency, an *ancillary statistic* $U = U(X^n)$ is one whose sampling dist'n doesn't depend on θ
- Similar to pivot, but U is a statistic
- If the distribution doesn't depend on θ , then U carries no (directly relevant) information about θ
- Why do we care about ancillary statistics?
- The word “ancillary” means *providing help*...
- It's not always easy/possible to construct ancillary statistics, but it works for structured (e.g., location–scale) models

- A statistic T is *complete* if

$$E_{\theta}\{f(T)\} = 0 \text{ for all } \theta \iff f \equiv 0$$

- In words, T is complete if no non-constant functions of T are ancillary statistics
- Finding complete statistics isn't trivial; see below
- Key properties:
 - roughly, a complete suff stat is a perfect data summary
 - complete + sufficient implies minimal sufficient
 - reg exp fam's admit a complete suff stat, most others don't
 - relevant for optimal estimation
 - *Basu's theorem*: if T is complete + suff and U is ancillary, then T and U are independent random variable

Complete statistics, cont.

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$, then $\sum_{i=1}^n X_i$ is complete

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Pois}(\theta)$, then $\sum_{i=1}^n X_i$ is complete

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\theta)$, then $\sum_{i=1}^n X_i$ is complete

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$, then $\max X_i$ is complete

- Roughly, the minimal sufficient statistic satisfies

$$T \longleftrightarrow (\underbrace{\text{an estimator of } \theta}_{\text{dimension of parameter}}, \text{ancillary statistic})$$

- If T is complete, then the ancillary component drops off
- Otherwise, when we reduce T further to, e.g., an estimator of θ , then we're losing information
- There's a way to protect from loss of information...
- But first we need to know what "information" means

Regularity conditions.

- 1 $\theta \mapsto p_\theta(x)$ is differentiable for each x
- 2 integration wrt x and differentiation wrt θ can be interchanged
- 3 $\text{supp}(P_\theta) := \{x : p_\theta(x) > 0\}$ doesn't depend on θ

- Score function: for iid's X^n

$$S_\theta(X^n) = \sum_{i=1}^n S_\theta(X_i) = \sum_{i=1}^n \frac{\partial}{\partial \theta} \log p_\theta(X_i) \quad \leftarrow \text{dim of } \theta$$

- Key property: $E_\theta\{S_\theta(X^n)\} = 0$
- Fisher information is the variance of the score...

Definition – Fisher information.

$$I_{X^n}(\theta) = V_{\theta}\{S_{\theta}(X^n)\} = E_{\theta}\{S_{\theta}(X^n) S_{\theta}(X^n)^{\top}\}$$

- Not too difficult to evaluate
- Shortcut: $I_{X^n}(\theta) = nI_{X_1}(\theta)$, see below
- Most problems have additional smoothness and, in such cases, there's an alternative formula

$$I_{X^n}(\theta) = -E_{\theta}\left\{\frac{\partial}{\partial\theta} S_{\theta}(X^n)\right\} = -E_{\theta}\left\{\frac{\partial^2}{\partial\theta^2} \log p_{\theta}(X^n)\right\}$$

- This is often more convenient for computations

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$, then $I_{X^n}(\theta) = \frac{n}{\theta(1-\theta)}$

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Pois}(\theta)$, then $I_{X^n}(\theta) = \frac{n}{\theta}$

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\theta)$, then $I_{X^n}(\theta) = \frac{n}{\theta^2}$

Exercise.

If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Gam}(\alpha, \beta)$, with $\theta = (\alpha, \beta)$, then

$$I_{X^n}(\theta) = \dots$$

- Interpretation of $I_{X^n}(\theta)$ as a measure of information comes from the mathematical properties it satisfies:
 - $I_{X^n}(\theta)$ is positive semi-definite (info's non-negative)
 - $I_{X^n}(\theta) = nI_{X_1}(\theta)$ (info accumulates)
- The main result explains:
 - why $I_{X^n}(\theta)$ measures “information”
 - why a sufficient stat $T(X^n)$ retains all the information in X^n

Theorem.

If $T(X^n)$ is a statistic, then

$$I_{X^n}(\theta) \geq I_{T(X^n)}(\theta),$$

with equality iff T is sufficient

- Intuitively, more informative data should permit higher quality estimation of unknowns
- The *Cramer–Rao lower bound* makes this intuition precise
- It gives a bound on the variance of unbiased⁶ estimators

Theorem.

Under the above regularity conditions, if $\phi : \mathbb{T} \rightarrow \mathbb{R}$ is smooth, then any unbiased estimator $\hat{\phi}_n$ of $\phi(\theta)$ satisfies

$$V_{\theta}(\hat{\phi}_n) \geq \dot{\phi}(\theta)^{\top} I_{X^n}(\theta)^{-1} \dot{\phi}(\theta)$$

Proof.

Exercise □

⁶An estimator $\hat{\phi}_n$ of $\phi(\theta)$ is unbiased if $E_{\theta}(\hat{\phi}_n) = \phi(\theta)$ for all θ

Exercise.

For iid $\text{Pois}(\theta)$, find the CR bound for estimating $\phi(\theta) = \theta^2$

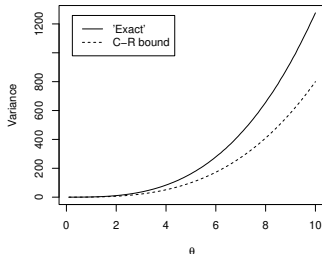
Exercise.

For iid $N(\mu, \sigma^2)$ data, with $\theta = (\mu, \sigma^2)$, find the CR bound for estimating $\phi(\theta) = \mu\sigma^2$

Exercise.

The lower bound is attained iff $S_\theta(X^n) = c(\theta)\{\hat{\phi}(X^n) - \phi(\theta)\}$

- The condition for equality in the CR bound is restrictive, so it's uncommon for the bound to be attained
- For example, with iid $\text{Pois}(\theta)$ data:
 - the bound can be attained for $\phi(\theta) = \theta$
 - but not for $\phi(\theta) = \theta^2$
 - plot shows variance of the “best” unbiased est of θ^2



Illustrative example.

Suppose that X_1, X_2 are iid with mass function

$$P_\theta(X = \theta - 1) = P_\theta(X = \theta) = P_\theta(X = \theta + 1) = \frac{1}{3}$$

Points to note:

- order stat's $(X_{(1)}, X_{(2)})$ are minimal sufficient
- $\hat{\theta} = \frac{1}{2}(X_{(1)} + X_{(2)})$ is an unbiased estimator
- $U = X_{(2)} - X_{(1)}$ is ancillary
- $(\hat{\theta}, U)$ is a one-to-one function of $(X_{(1)}, X_{(2)})$

U isn't directly relevant, but ignoring it means a loss of info. It is indirectly relevant: e.g., if $U = 2$, then $\hat{\theta}$ is exactly θ

- The above situation isn't uncommon
- When the min suff stat T has higher dim than θ
 - any estimator that's a function of T loses info
 - incorporating an ancillary U can help, in a specific sense, to recover the info that would be lost
- How to incorporate U ? *Conditioning!*
- If $T = (\hat{\theta}, U)$, then work with the conditional sampling distribution of $\hat{\theta}$, given $U = u$
- e.g., the relevant conditional information is

$$I_{\hat{\theta}|U}(\theta | u) = V_{\theta} \underbrace{\left\{ \frac{\partial}{\partial \theta} \log p_{\theta}(\hat{\theta} | u) \right\}}_{\text{conditional score}}$$

Theorem.

Conditioning recovers would-be-lost info, i.e., on average the conditional information equals the full information in the sample

$$\underbrace{E\{ I_{\hat{\theta}|U}(\theta | U) \}}_{\text{conditional info}} = \underbrace{I_{X^n}(\theta)}_{\text{total info}} > \underbrace{I_{\hat{\theta}}(\theta)}_{\text{marginal info}}$$

- There are complications in carrying this out⁷
- Typically $I_{X^n}(\theta) - I_{\hat{\theta}}(\theta) \rightarrow 0$ as $n \rightarrow \infty$
- But skin-in-the-game dictates that we do the best we can, i.e., we shouldn't dismiss conditioning out of hand

⁷See, e.g., Brazzale, Davison, and Reid's *Applied Asymptotics*

Cox's measurement device example.

Data consists of $X = (\hat{\theta}, U)$, where

$$(\hat{\theta} \mid U = u) \sim \begin{cases} \text{N}(\theta, 1) & \text{if } u = 1 \\ \text{N}(\theta, \frac{1}{99}) & \text{if } u = 0 \end{cases} \quad \text{and} \quad U \sim \text{Ber}(\frac{1}{2})$$

Of course, U is ancillary, but we lose information if we ignore it

$$\frac{1}{2}(1 + \frac{1}{99}) = I_X(\theta) > I_{\hat{\theta}}(\theta) = \text{hard to calculate...}$$

Conditional info is easy

$$I_{\hat{\theta}|U}(\theta \mid u = 1) = 1 \quad \text{and} \quad I_{\hat{\theta}|U}(\theta \mid u = 0) = 99$$

And easy to check that $E\{I_{\hat{\theta}|U}(\theta \mid U)\} = I_X(\theta)$

- We're often only interested in features of θ
 - $\phi = \phi(\theta)$ is the interest parameter
 - $\lambda = (\text{everything else})$ is a nuisance parameter
- Tempting to want to find stats “sufficient for ϕ ”
 - this generally doesn't work
 - but sometimes the model factors nicely...
- I'll skip it here, but see Section 3.7 of my draft book

- Two statistical principles (roughly):
 - SP. Inference depends on data only via a min-suff stat
 - CP. Inference should condition on ancillary statistics
- To me, CP is just “common sense” (see Cox’s example)
- These aren’t too controversial...
- But there’s another, the *likelihood principle*, that’s
 - more controversial
 - shown to be a consequence of SP + CP
- Lots of standard statistical methods violate LP
- So then they must violate at least one of the not-too-controversial principles, which is troubling...

Questions, etc?

- I haven't talked yet about *how* to construct good estimators, p-value functions, etc
- Reason: we've yet to see the key ingredient, the *likelihood function*, which drives most of modern statistics

Definition.

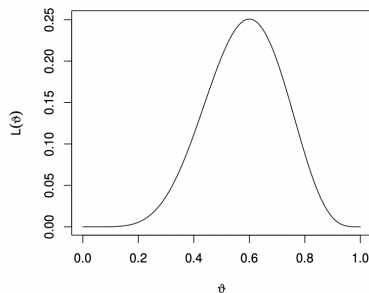
If X^n is an iid sample from P_θ , then the *likelihood function* for θ is

$$L_n(\vartheta) = L_{X^n}(\vartheta) = p_\vartheta(X^n) = \prod_{i=1}^n p_\vartheta(X_i), \quad \vartheta \in \mathbb{T}$$

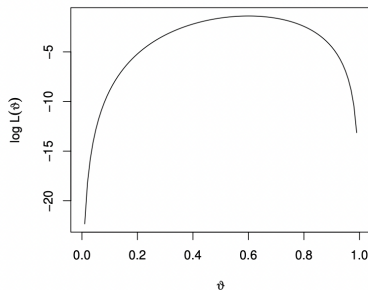
The log-likelihood is $\ell_n(\vartheta) = \ell_{X^n}(\vartheta) = \log L_n(\vartheta)$

- What we're going to do with likelihood is almost “common sense,” so it's a very old idea
- But Fisher was the first to formalize this and establish it's fundamental role in statistical theory
- Roughly, the likelihood function ranks the parameter values in terms of how well they fit the data X^n
- likelihood $\neq$ probability!

[The likelihood] is not however a probability and does not obey the laws of probability... it does none the less afford a rational basis for preferring some values of θ ... to others. —Fisher



(a) Likelihood



(b) Log-likelihood

Figure 4.1: Likelihood and log-likelihood functions based on binomial model with observed $X = 6$ and $n = 10$; the corresponding sample proportion is $\hat{\theta} = 0.6$.

- Recall: likelihood ranks the parameter values
- Natural to seek the “highest ranked” parameter value, which corresponds to the value that best fits the data
- That “highest ranked” parameter value corresponds to maximizing the (log) likelihood function

Definition.

A *maximum likelihood estimator* (MLE), if it exists, is

$$\hat{\theta}_n \in \arg \max_{\vartheta} L_n(\vartheta)$$

Can replace likelihood with log-likelihood in the above display

Exercise.

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$; find the MLE

Exercise.

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Pois}(\theta)$; find the MLE

Exercise.

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$; find the MLE

Exercise.

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{N}(\mu, \sigma^2)$, with $\theta = (\mu, \sigma^2)$; find the MLE

- When ℓ_n is smooth, optimization becomes root-finding:
 - $\dot{\ell}_n(\theta) = 0$ is the *likelihood equation*
 - MLE solves the likelihood equation
- Technically, to confirm that the solution is a maximum, need to check a second-order condition
 - $\ddot{\ell}_n(\hat{\theta}_n)$ is negative definite
 - $J_n(\hat{\theta}_n) = -\ddot{\ell}_n(\hat{\theta}_n)$ is the *observed Fisher information*
- Typically, numerical methods are needed to get the MLE
 - Newton's method from calculus
 - generic optimization routines like `optim` in R

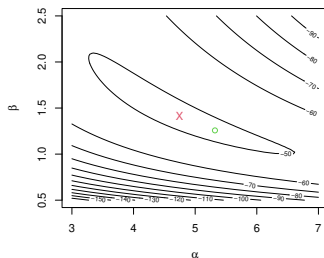
Maximum likelihood, cont.

Example.

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Gam}(\alpha, \beta)$ where $\theta = (\alpha, \beta)$. We can write out the likelihood equation, but can't solve it in closed form

$$\dot{\ell}_n(a, b) = \begin{pmatrix} -n \log b - n\psi(a) + \sum_{i=1}^n \log X_i \\ -\frac{na}{b} + \frac{1}{b^2} \sum_{i=1}^n X_i \end{pmatrix} = 0$$

But numerical methods are easy and readily available



Exercise.

Let $X_1, \dots, X_n$ be iid from a one-parameter exponential family

$$p_{\theta}(x) = h(x) \exp\{\eta(\theta) t(x) - B(\theta)\}$$

Show that the likelihood equation is equivalent to

$$\frac{1}{n} \sum_{i=1}^n t(X_i) = \frac{\dot{B}(\vartheta)}{\dot{\eta}(\vartheta)} = \mathbb{E}_{\vartheta}\{t(X_1)\}$$

- MLE existence and uniqueness isn't automatic
- Statistical properties discussed later
- MLEs can easily accommodate non-trivial constraints:
 - suppose we *know* that $\theta \in \mathbb{T}_0 \subset \mathbb{T}$
 - get the constrained MLE, $\hat{\theta}_n^0 \in \arg \max_{\theta \in \mathbb{T}_0} \ell_n(\theta)$
- The MLE also satisfies an invariance property
 - MLE of a function is the function of the MLE
 - really convenient/useful property

Theorem.

If $\hat{\theta}_n$ is the MLE of θ , then $\hat{\phi}_n := \phi(\hat{\theta}_n)$ is the MLE of $\phi = \phi(\theta)$

Exercise.

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$; find the MLE of $V_\theta(X_1)$

Exercise.

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Pois}(\theta)$; find the MLE of $P_\theta(X_1 = 0)$

Exercise.

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, with $\theta = (\mu, \sigma^2)$; find the MLE of $E_\theta(X_1^2)$

Exercise.

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Gam}(\alpha, \beta)$, with $\theta = (\alpha, \beta)$; find the MLE of $E_\theta(X_1)$

- The likelihood function's magnitude is meaningless, all that matters is the shape/rankings
- So we could rescale the likelihood to $[0, 1]$
- The *relative likelihood* is

$$R_n(\vartheta) = R(X^n, \vartheta) = \frac{L_n(\vartheta)}{L_n(\hat{\theta}_n)}, \quad \vartheta \in \mathbb{T}$$

- R_n gives the same rankings as L_n and ℓ_n
- Key properties:
 - the function $\vartheta \mapsto R_n(\vartheta)$ is minimal sufficient
 - random variable $R(X^n, \theta)$ is an approx pivot — later

Exercise.

For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$, find the relative likelihood

Exercise.

For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\theta)$, find the relative likelihood

Exercise.

For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$, find the relative likelihood

Exercise.

For $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$, find the relative likelihood

- I haven't told you how to construct a p-value yet
- That's because we didn't have the key ingredient
- *My proposal*: construct p-value using relative likelihood
- Intuition: ϑ and x^n are compatible when $R(x^n, \vartheta)$ is large
- Suggests taking $T_{\vartheta}(x^n) = 1/R(x^n, \vartheta)$
- Specifically, my proposal is

$$\pi_{x^n}(\vartheta) = P_{\vartheta}\{R(X^n, \vartheta) \leq R(x^n, \vartheta)\}, \quad \vartheta \in \mathbb{T}$$

- The p-value plots I showed before are based on this
- Why is this reasonable/good?
 - natural measure of compatibility
 - makes efficient use of data (min suff stat)
 - relative likelihood is an approx pivot
 - limit distribution is known $\rightsquigarrow$ simple approximation
- The general things I said before about p-values apply here
 - $\pi_{X^n}(\theta)$ is (no smaller than) $\text{Unif}(0, 1)$ under P_θ
 - can derive tests & confidence sets from this p-value
 - these inherit error rate control, key for inductive logic

- For a simple null hypothesis $H = \{\theta_0\}$, the (level- α) *likelihood ratio test* is equivalent to

reject hypothesis “ $\theta = \theta_0$ ” iff $\pi_{x^n}(\theta_0) \leq \alpha$

- Inverting the above test gives a $100(1 - \alpha)\%$ likelihood-based confidence set

$$C_\alpha(x^n) = \{\vartheta : \pi_{x^n}(\vartheta) > \alpha\}$$

- These are the same test & confidence procedures I described in the context of a general p-value function

Exercise.

For iid $N(\theta, 1)$ data, characterize the event " $\pi_{x^n}(\theta_0) \leq \alpha$ "

Exercise.

For iid $\text{Unif}(0, \theta)$ data, characterize the event " $\pi_{x^n}(\theta_0) \leq \alpha$ "

Exercise.

For iid $\text{Exp}(\theta)$ data, characterize the event " $\pi_{x^n}(\theta_0) \leq \alpha$ "

- The general p-value formulation I presented before still applies when the hypothesis isn't simple
- But there's a different (and sometimes better) test that is commonly referred to as the *likelihood ratio test*
- Start by extending the relative likelihood to set arguments:

$$R_n(H) = \sup_{\vartheta \in H} R_n(\vartheta), \quad H \subseteq \mathbb{T}$$

- Intuition: don't reject if any $\vartheta \in H$ is compatible with X^n
- Likelihood ratio test rejects " $\theta \in H$ " iff $R_n(H)$ is small
- Threshold depends on the desired level α and other features of the model; often an asymptotic approx is used

Likelihood: tests & confidence sets, cont.

Example.

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$, so that $T := \sum_{i=1}^n X_i$ is $\text{Bin}(n, \theta)$. Suppose the goal is to test if $\theta \in H = [0, \theta_0]$. Below is a plot of the likelihood ratio statistic $R_n(H)$ as a function of T .

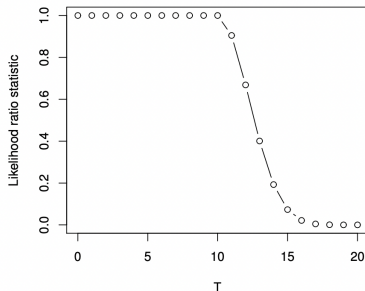


Figure 4.6: Plot of the likelihood ratio statistic, as a function of the number of successes T , for the null hypothesis $H : \theta \leq \theta_0$, with $n = 20$ and $\theta_0 = 0.5$.

Exercise.

$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, with $\theta = (\mu, \sigma^2)$ unknown. Find the LR test of “ $\mu = \mu_0$ ” or, equivalently, of $\theta \in H := \{(\mu, \sigma^2) : \mu = \mu_0\}$

Exercise.

Data consists of two independent samples

$$X_1^{n_1} = \underbrace{(X_{11}, \dots, X_{1n_1})}_{\stackrel{\text{iid}}{\sim} N(\mu_1, \sigma_1^2)} \quad \text{and} \quad X_2^{n_2} = \underbrace{(X_{21}, \dots, X_{2n_2})}_{\stackrel{\text{iid}}{\sim} N(\mu_2, \sigma_2^2)},$$

Assuming σ_1^2 and σ_2^2 are equal to a known σ^2 , find the LR test of the hypothesis that the group means are equal, i.e., $\mu_1 = \mu_2$.

- For likelihood solutions to be “good,” we need to ensure that they’re compatible with our inductive logic
- That is, giving a wrong answer should be improbable in some specific sense
- A first sense is consistency of the MLE
- This is non-trivial because typically we need numerical methods to find the MLE , i.e., no closed-form
- But there are general consistency theorems...

- Start with something relatively simple...
- Scalar exponential family model with PDF/PMF:

$$p_{\theta}(x) = h(x) \exp\{\eta(\theta) t(x) - B(\theta)\}, \quad \theta \in \mathbb{R}$$

- Log-likelihood function is

$$\ell_n(\vartheta) = \eta(\vartheta) \sum_{i=1}^n t(X_i) - nB(\vartheta) + (\text{constant in } \vartheta)$$

- Likelihood equation:

$$\dot{\ell}_n(\vartheta) = 0 \iff \frac{1}{n} \sum_{i=1}^n t(X_i) = \frac{\dot{B}(\vartheta)}{\dot{\eta}(\vartheta)}$$

Exercise.

- $m(\theta) := E_{\theta}\{t(X_1)\} = \dot{B}(\theta)/\dot{\eta}(\theta)$
- $v(\theta) := V_{\theta}\{t(X_1)\} = \dot{m}(\theta) = \dots \geq 0$
- Likelihood equation: $n^{-1} \sum_{i=1}^n t(X_i) = m(\vartheta)$
- Function m is increasing, so there's a solution

$$\hat{\theta}_n = m^{-1}\left(\frac{1}{n} \sum_{i=1}^n t(X_i)\right)$$

- Two observations:
 - m^{-1} is continuous
 - $n^{-1} \sum_{i=1}^n t(X_i) \rightarrow m(\theta)$ by LLN
- Continuous mapping theorem implies $\hat{\theta}_n$ is consistent!

- The above result/example relies heavily on the nice structure of exponential family models
- Can't expect things to be so easy more generally
- I like the following result:
 - not a consistency theorem
 - strongly suggests that we should expect consistency
 - i.e., $L_n(\theta)$ tends to be higher than any $L_n(\vartheta)$

Theorem.

Let $X^n = (X_1, \dots, X_n)$ be iid P_θ , with likelihood L_n . Then for any pair of distinct parameter values (θ, ϑ) , the following holds:

$$P_\theta\{L_n(\theta) > L_n(\vartheta)\} \rightarrow 1, \quad n \rightarrow \infty$$

- A general consistency theorem for MLEs is available, but is rather technical; see Theorem 4.2 in the draft book
- The point is that it's not enough for the likelihood to eventually be larger at the true θ than at any other fixed ϑ
- This ordering needs to hold uniformly over sets of ϑ 's
- Suffice it to say, this often holds, just not easy to check
- See Example 4.26 in the draft book, general location models
- *It can happen that the MLE is inconsistent* — later

- Beyond consistency, we can ask about convergence rates and limit distributions for MLEs
- These details are relatively easy provided that
 - consistency holds
 - problem is sufficiently regular
- The take-away message is that
 - MLE is $n^{-1/2}$ -consistent
 - MLE is asymptotically normal
 - MLE is asymptotically efficient in the sense that the variance of the limiting normal matches the Cramer–Rao bound

Regularity conditions.

- 1 Support of P_θ doesn't depend on θ
- 2 $\theta \mapsto p_\theta(x)$ is twice continuously differentiable for all x
- 3 Derivatives & integrals can be interchanged, hence the score function has mean zero and the Fisher information exists and can be evaluated using two equivalent formulas
- 4 For all small $r > 0$, there exists functions H_r such that, for each θ and all x ,

$$\max_{j,k} \sup_{\|\vartheta - \theta\| \leq r} \left| \frac{\partial^2 \log p_\vartheta(x)}{\partial \vartheta_j \partial \vartheta_k} - \frac{\partial^2 \log p_\theta(x)}{\partial \vartheta_j \partial \vartheta_k} \right|_{\vartheta=\theta} \leq H_r(x, \theta)$$

$$\lim_{r \rightarrow 0} E_\theta \{H_r(\theta, X_1)\} = 0.$$

Theorem.

Under above regularity conditions, for any θ in the interior of $\mathbb{T}$ with Fisher information $I(\theta) > 0$, if $\hat{\theta}_n$ is a consistent solution of the likelihood equation, then

$$n^{1/2}(\hat{\theta}_n - \theta) \rightarrow N(0, I(\theta)^{-1}) \quad \text{in distribution}$$

- Above scaling implies $n^{-1/2}$ -consistency
- Asymptotic variance agrees with Cramer–Rao bound
- Extends to the MLE of any suitable transformation $\phi(\theta)$ of θ by invariance and the delta theorem, i.e.,

$$n^{1/2} \left\{ \underbrace{\phi(\hat{\theta}_n)}_{\text{MLE of } \phi(\theta)} - \phi(\theta) \right\} \rightarrow N\left(0, \underbrace{\dot{\phi}(\theta)^\top I(\theta)^{-1} \dot{\phi}(\theta)}_{=v^\phi(\theta)}\right)$$

- Several asymptotic pivots emerge from the above theorem
- The most obvious is the Wald pivot:

$$\frac{n^{1/2}\{\hat{\phi}_n - \phi(\theta)\}}{v^{\phi}(\theta)^{1/2}} \rightarrow N(0, 1)$$

- Wald's asymptotic $100(1 - \alpha)\%$ conf. interval for $\phi(\theta)$

$$\hat{\phi}_n \pm \text{qnorm}(1 - \frac{\alpha}{2})\{v^{\phi}(\hat{\theta}_n)/n\}^{1/2}$$

- Other options for variance besides simple plug-in:
 - observed Fisher information⁸
 - bootstrap

⁸This has some advantages, which I can discuss if there's time...

- My favorite pivot is the relative likelihood
- Under the same conditions as the MLE asymptotic normality theorem, it follows that $-2 \log R_n(\theta)$ has a chi-square limit
- Leads to a simple p-value function approximation

Wilks's theorem, part I.

Under the conditions of the previous theorem,

$$-2 \log R_n(\theta) \rightarrow \text{Chisq}(\nu), \quad \text{in distribution}$$

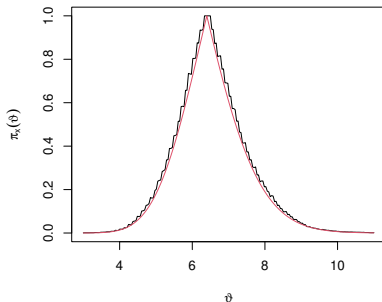
where ν is the dimension of θ

- P-value function approximation

$$\begin{aligned}\pi_{x^n}(\vartheta) &= P_{\vartheta}\{R(X^n, \vartheta) \leq R(x^n, \vartheta)\} \\ &= P_{\vartheta}\{-2 \log R(X^n, \vartheta) \geq -2 \log R(x^n, \vartheta)\} \\ &\approx 1 - \underbrace{G_{\nu}(-2 \log R(x^n, \vartheta))}_{\text{Chisq}(\nu) \text{ CDF}}\end{aligned}$$

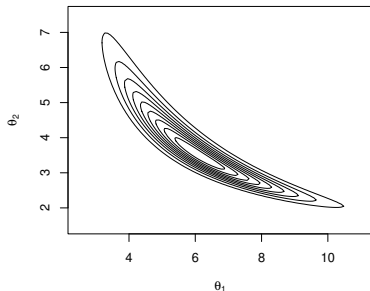
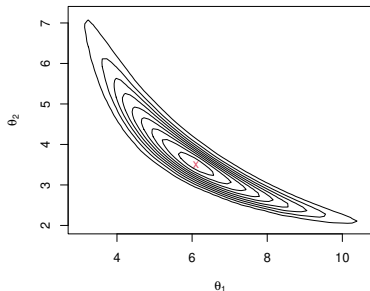
- Easy to construct tests and confidence sets as before based on this approximate p-value function
- Approximately valid and efficient
- These tests & confidence sets agree with what is commonly used in practice, e.g., the `glm` function in R

- Example...
- $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Pois}(\theta)$
- P-value function
 - exact
 - Wilks's approx
- Plot: $n = 5, \bar{x} = 7$



Likelihood asymptotics, cont.

- $\text{Gam}(\theta_1, \theta_2)$ model
- Exact p-value (left), Wilks's approximation (right)
- Based on $n = 25$ samples from $\text{Gam}(7, 3)$



- Limit distribution of extended relative likelihood?
- Suppose H imposes M -many constraints on $\theta \in \mathbb{R}^D$

$$H : g_1(\theta) = \cdots = g_M(\theta) = 0, \quad M \leq D$$

- Common examples:
 - sets the values of M components, $g_m(\theta) = \theta_m - \theta_{m0}$
 - sets two components equal, $g_1(\theta) = \theta_2 - \theta_1$

Wilks's theorem, part II.

Under the conditions of the previous theorem, for H as above,

$$-2 \log R_n(H) \rightarrow \text{Chisq}(M), \quad \text{in distribution}$$

Exercise.

Data consists of two independent samples

$$X_1^{n_1} = (\underbrace{X_{11}, \dots, X_{1n_1}}_{\text{iid } \text{Pois}(\theta_1)}) \quad \text{and} \quad X_2^{n_2} = (\underbrace{X_{21}, \dots, X_{2n_2}}_{\text{iid } \text{Pois}(\theta_2)}),$$

Find the approximate LR test of the hypothesis “ $\theta_1 = \theta_2$ ”

Exercise.

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$. Show that Wilks's theorem doesn't apply for testing the hypothesis “ $\theta \leq \theta_0$ ”

- To summarize:
 - MLEs are consistent, asymptotically normal + efficient
 - relative likelihood is an asymptotic pivot
 - $-2 \log R_n(\cdot)$ has a chi-square limiting distribution
- Of course, the above statements involve conditions
- Conditions can be checked in exponential families
- But there are still some subtleties...

- There are cases when likelihood methods fall short
- For example, sometimes
 - MLE doesn't exist
 - MLE isn't unique
 - MLE is inconsistent
 - MLE isn't strictly efficient
 - relative likelihood-based methods fail to be valid
 - ...
- These aren't counterexamples to the theorems, they just highlight the subtleties lurking
- Not a criticism of the likelihood-based methods, just a warning that you shouldn't blindly use them

- Famous example of inconsistency: Neyman–Scott problem
 - independent pairs $X_i = (X_{i1}, X_{i2})$, within are iid $N(\mu_i, \sigma^2)$
 - interest is in σ^2
 - $\hat{\sigma}_n^2 = n^{-1} \sum_{i=1}^n (X_{i1} - X_{i2})^2$
 - converges in probability to $\sigma^2/2 \neq \sigma^2$
- It's easy to adjust the estimator so that it's consistent,⁹ but that's not the point
- The point is that ML *can* be inconsistent
- There are important insights in this example...
 - I'll sketch this out on the board
 - using a more modern regression context

⁹Can also modify the definition of likelihood, so that the corresponding modified MLE is consistent

- Fisher's claim was that MLEs were the best, but...
- Hodges's famous example:
 - $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$, MLE is $\hat{\theta}_n = \bar{X}_n$
 - define the new estimator

$$\tilde{\theta}_n = \begin{cases} \hat{\theta}_n & \text{if } |\hat{\theta}_n| \geq n^{-1/4} \\ c\hat{\theta}_n & \text{if } |\hat{\theta}_n| < n^{-1/4}, \end{cases} \quad c \in (-1, 1)$$

- can be shown that

$$n^{1/2}(\tilde{\theta}_n - \theta) \rightarrow \begin{cases} N(0, 1) & \text{if } \theta \neq 0 \\ N(0, c^2) & \text{if } \theta = 0 \end{cases}$$

- $\tilde{\theta}_n$ is *super-efficient* because it can have variance smaller than the Cramer–Rao bound
- Super-efficiency can occur only in *small* subsets of $\mathbb{T}$

Warnings, cont.

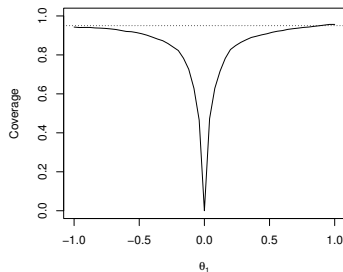
- *Fieller–Creasy problem*:

- $X_1^n \stackrel{\text{iid}}{\sim} N(\theta_1, 1)$ and $X_2^n \stackrel{\text{iid}}{\sim} N(\theta_2, 1)$
- goal is inference on $\phi = \theta_2/\theta_1$

- Simple asymptotic $100(1 - \alpha)\%$ confidence interval

$$\hat{\phi}_n \pm z_{\alpha/2} (n^{-1} \hat{v}_n^\phi)^{1/2}, \quad \hat{v}_n^\phi = \frac{\hat{\theta}_1^2 + \hat{\theta}_2^2}{\hat{\theta}_1^4}$$

- But its coverage probability can be arbitrarily low...



- Likelihood has nice intuition
- Likelihood-based methods are theoretically sound
- Numerical optimization is often needed, but that's OK
- Strong asymptotic properties
- But there are subtleties, can't use methods blindly
- One foundational issue:
 - LP. inference only depends on observed likelihood function
 - Birnbaum: $SP + CP \implies LP$
 - even likelihood-based methods don't satisfy LP
 - so they must violate $CP \longleftrightarrow$ "common sense"

Exercise.

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Gam}(\alpha, \theta)$ with α known. Find the MLE

Exercise.

MLEs are generally biased. For example, find the MLE of $\phi(\theta) = \theta(1 - \theta)$ in the iid Bernoulli model and calculate the bias. Show, however, that the bias is vanishing as $n \rightarrow \infty$

Exercise.

Exercise 4.32 in my draft book — for homework

Questions, etc?

- Bayesian methods
 - perspective
 - prior-to-posterior update
 - estimators, tests, etc
 - examples
 - foundational benefits
 - choice of prior distribution
 - asymptotic properties

- Bayesian inference is very attractive
 - all boils down to applied probability
 - in particular, fully probabilistic UQ
 - it's clear how to solve any new problem
- For these and perhaps other reasons, at one time I might have considered myself a Bayesian
- But now I'm not — the benefits come at a high price
- I have a unique perspective to share, since it's usually only Bayesians that talk about Bayes...
- My goal here:
 - describe how Bayes works and what's good about it
 - highlight the issues (as I see them)

- So far, I've not explained what " θ unknown" means
- No explanation was needed for what we were doing
- Bayesians insist on a probabilistic explanation
- Bayesian inference is based on the following "axiom"
all uncertainties are described using probabilities
- Explain " θ unknown" in terms of probabilities?
- Not so straightforward...
- Thought experiment: θ = mean income in my hometown

- Simple illustration: cancer screening
 - data $T = \{\text{test positive}\}$
 - hypothesis $H = \{\text{has cancer}\}$
 - model: $P(T | H) = 0.95$ and $P(T | H^c) = 0.05$
 - prior $P(H) = 0.01$
- *Question:* patient tests positive — cancer or not?
- Polarizing example
 - Bayesians infer H^c
 - non-Bayesians infer H

- Non-Bayesian solution:

- *law of likelihood*

$$P(T \mid H) \div P(T \mid H^c) = 19 \gg 1 \implies \text{infer } H$$

- p-value, $\pi_T(H) = 1 \implies \text{infer } H$

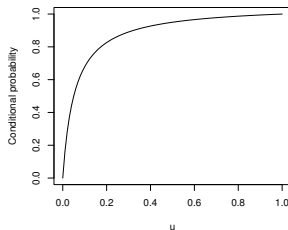
- Bayesian solution:

- Bayes formula $P(H) \rightarrow P(H \mid T)$,

$$u \mapsto \frac{0.95 u}{0.95 u + 0.05 (1 - u)}$$

- if $u = 0.01$, then $P(H \mid T) = 0.16$

- $0.16 \ll \frac{1}{2} \implies \text{infer } H^c$



I would like to object to the statement... that a prior is unknown. This is ridiculous, a prior is a statement of one's knowledge and modern work demonstrates that it is always known: by judicious questioning it can be found. —Lindley

- There's nothing wrong with genuine/subjective prior info
- But often genuine prior info is not available¹⁰
- If a prior is required, then the choice must matter
- And there must be consequences somewhere for choosing a fake rather than genuine prior

¹⁰e.g., “Scientists like to work on new problems” (Efron)

[Bayes's formula] does not create real probabilities from hypothetical probabilities. —Fraser

- If prior isn't real, then posterior isn't "real"
 - not clear then how Bayesian's UQ is meaningful
 - posterior probabilities might not be reliable
 - i.e., false hypotheses tending to be assigned high posterior prob
- But Bayesian methods can still "work" in the same sense that likelihood methods work¹¹
 - sampling distribution properties
 - e.g., consistent estimators
- Solution is "right" versus the method "works"

¹¹See Fraser (Stat Sci 2011), *Is Bayes just quick and dirty confidence?*

- In this Bayesian context, write Θ for the uncertain parameter
- The Bayesian model has two pieces:
 - prior $\Theta \sim Q$ with PDF q
 - model $(X_1, \dots, X_n \mid \Theta = \theta) \stackrel{\text{iid}}{\sim} P_\theta$
- Implies a joint dist'n for (X^n, Θ)
- The posterior distribution Q_n is the corresponding conditional dist'n of Θ , given $X^n = x^n$, with PDF

$$q_n(\theta) = \frac{p_\theta(x^n) q(\theta)}{\int p_\vartheta(x^n) q(\vartheta) d\vartheta} \propto \text{likelihood} \times \text{prior}$$

- Q_n depends on data only through a minimal sufficient stat

- This is beautiful, now we have literally everything we need
- Posterior Q_n contains all the relevant info to answer questions about uncertain Θ , given $X^n = x^n$
- Things we can do with Q_n :
 - estimation
 - credible sets
 - hypothesis testing
 - ...
- This all boils down to “summarizing” the posterior
- I'll start with general strategies, then we'll look at examples where we can put them into action

- Bayes estimator of Θ is the posterior mean, which depends on data through the posterior q_n :

$$\hat{\theta}_n = \int \theta q_n(\theta) d\theta$$

- More generally, Bayes estimator of $\phi(\theta)$ is

$$\hat{\phi}_n = \int \phi(\theta) q_n(\theta) d\theta$$

- Further justification for choosing the mean (as opposed to, say, median) of the posterior is given later

- Set estimator for Θ ?
- A $100(1 - \alpha)\%$ credible set is a (data-dependent) subset $C_\alpha(x^n)$ of the parameter space such that

$$Q_n\{\Theta \in C_\alpha(x^n)\} \geq 1 - \alpha$$

- Not unique, there are a couple of standard strategies:
 - for scalar Θ , find lower & upper posterior quantiles
 - for general Θ , find the *highest posterior density* region
- Better to show these in the context of examples...
- But note: *a credible set is generally not a confidence set*

- Hypothesis " $\Theta \in H$ "
- If $Q(H) > 0$, then things are relatively easy
- Bayes test¹² rejects " $\Theta \in H$ " iff $Q_n(H) < \frac{1}{2}$
- But it's often the case that $Q(H) = 0$, e.g., $H = \{\theta_0\}$, and this creates serious complications
- Fact:¹³ if $Q(H) = 0$, then $Q_n(H) = 0$, no matter the data
- Two ways Bayesians sidestep this issue:
 - choose a prior that incorporates H
 - use a Bayes factor
- I can explain these if there's time/interest

¹² Justification of the threshold $\frac{1}{2}$ comes later...

¹³ It's also true that if $Q(H)$ is small, then $Q_n(H)$ tends to be small too

Bayesian practice, cont.

Example.

- Bayesian model:

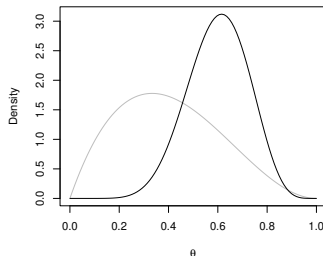
- $(X_1, \dots, X_n \mid \Theta = \theta) \stackrel{\text{iid}}{\sim} \text{Ber}(\theta)$

- $\Theta \sim \text{Beta}(a, b)$

- Posterior distribution is $Q_n = \text{Beta}(a_n, b_n)$, with

$$a_n = a + n\bar{X} \quad \text{and} \quad b_n = b + n - n\bar{X}$$

- Plot with $a = 2$, $b = 3$, $n = 10$, and $\bar{x} = 0.7$



Example, cont.

- Posterior mean is

$$\hat{\theta}_n = \frac{a_n}{a_n + b_n} = \frac{a + n\bar{X}}{a + b + n}$$

- 95% credible interval based on posterior CDF:

$$\text{qbeta}(c(0.025, 0.975), \text{shape1} = a_n, \text{shape2} = b_n)$$

- 95% highest posterior density interval/region
 - define $h(c) = Q_n(\{\theta : q_n(\theta) \geq c\})$
 - find $c = c(x^n)$ such that $h(c) = 0.95$
 - a picture will help clarify this...
 - generally “better” than the CDF-based solution, i.e., gives the narrowest interval with posterior prob 0.95

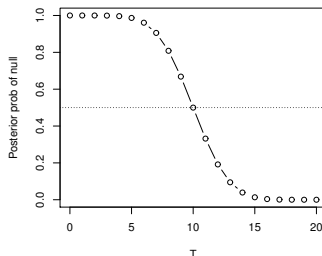
Bayesian practice, cont.

Example, cont.

- Let $H = [0, \theta_0]$, goal is to test “ $\Theta \in H$ ”
- Posterior probability of H is a CDF calculation

$$Q_n(H) = \text{pbeta}(\theta_0, \text{shape1} = a_n, \text{shape2} = b_n)$$

- Plot below shows $Q_n(H)$ as a function of $T = n\bar{X}$, with $n = 20$ and $\theta_0 = 0.5$



Exercise.

Find the posterior distribution of Θ if

- $(X_1, \dots, X_n \mid \Theta = \theta) \stackrel{\text{iid}}{\sim} \text{Pois}(\theta)$
- $\Theta \sim Q = \text{Gam}(a, b)$

Exercise.

Find the posterior distribution of Θ if

- $(X_1, \dots, X_n \mid \Theta = \theta) \stackrel{\text{iid}}{\sim} N(\theta, \sigma^2)$
- $\Theta \sim Q = N(m, v)$

Exercise.

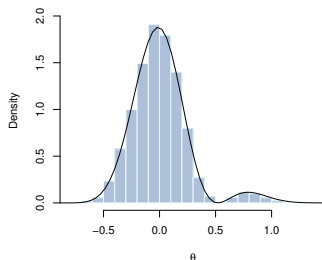
Find the posterior distribution of Θ if

- $(X_1, \dots, X_n \mid \Theta = \theta) \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$
- $\Theta \sim Q = \text{Pareto}(a, b)$, with $q(\theta) = \frac{ab^a}{\theta^{a+1}} 1(\theta \geq a)$

- The above examples all have one thing in common...
- The posterior has the same form as the prior
- These are called *conjugate priors*
- Until the late 1900s, these were virtually the only examples that could be done, due to computational limitations
- Restrictions on prior form are incompatible with the desired genuineness of prior info
- Without conjugacy, numerical methods are needed — in particular, Monte Carlo methods
- These technical advances rejuvenated Bayesian inference

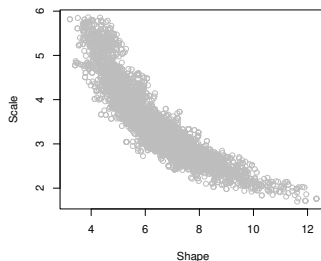
Example.

- Bayesian model:
 - $(X_1, \dots, X_n \mid \Theta = \theta) \stackrel{\text{iid}}{\sim} p_\theta(x) = (2\pi)^{-1} \{1 - \cos(x - \theta)\}$
 - $\Theta \sim \text{Unif}(-\pi, \pi)$
- Posterior has no nice form, but it can be evaluated
 - using numerical integration
 - using Markov chain Monte Carlo



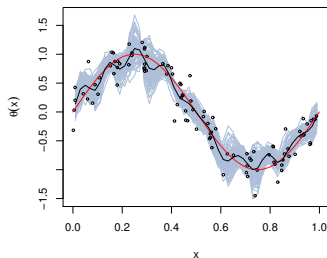
Example.

- Bayesian model:
 - $(X_1, \dots, X_n \mid \Theta = \theta) \stackrel{\text{iid}}{\sim} \text{Gam}(\alpha, \beta)$, with $\theta = (\alpha, \beta)$
 - $\Theta \sim [\text{Jeffreys}] \leftarrow$ to be defined
- Posterior has no nice form, but can use MCMC



Example.

- Bayesian (nonparametric) regression model:
 - $(Y_1, \dots, Y_n \mid \Theta = \theta) \stackrel{\text{ind}}{\sim} \mathcal{N}(\theta(x_i), \sigma^2)$
 - $\Theta \sim$ Gaussian process
- This model assumes Θ is a *random function*
- Monte Carlo methods can be used for very high-dim cases



- “ Θ is unknown” must be explained via probability
- Bayesians take a prior distribution Q and update to a posterior distribution Q_n using data X^n
- Answers to (most) questions are summaries of Q_n
- When the prior is nice, we can get the posterior in closed-form; if not, then we use numerical methods
- Later I'll show that the posterior has good stat properties
- Next I'll talk about some desirable foundational properties

- A sequence $X_1, X_2, \dots$ is *exchangeable* if, for each n , the joint dist'n of $(X_1, \dots, X_n)$ is invariant to permutations
- Examples
 - iid sequences
 - conditionally iid sequences
- In many applications, we'd feel comfortable saying that data are exchangeable, i.e., the order is irrelevant to inference
- De Finetti/Savage proved that exchangeability is equivalent to a Bayes (conditionally iid) model
- So, if the data are judged exchangeable, then
 - there exists a prior & likelihood that sets the joint dist'n
 - hence a Bayesian analysis is "right"

- Birnbaum: “SP + CP $\implies$ LP”
- The likelihood-based methods before violate LP
 - hence must violate SP and/or CP
 - such violations suggest a lack of common sense
- Bayesian methods,¹⁴ however, do satisfy LP
- This has major (philosophical) implications...

I, myself, came to take... Bayesian statistics... seriously only through recognition of the likelihood principle. —Savage

¹⁴The ones with a genuine, subjective prior

Desirable properties, cont.

- Probabilities are often understood as defining fair prices for relevant gambles
- i.e., probabilistic framework $\rightsquigarrow$ fair prices for gambles
- But it can go the other way too, start by defining a notion of “fair prices” and deriving a framework for UQ
- De Finetti showed a player can suffer *sure loss* unless his prices obey the laws of (finitely additive¹⁵) probabilities
- Statisticians are effectively taking gambles
 - if they reject a true hypothesis, then there is (should be) a loss
 - absurd if a stat. inference framework allowed for sure loss
- Bayesian formulation based fully on subjective probability is immune to sure loss, hence is *coherent*

¹⁵Kolmogorov's countable additivity has a role here too...

- Given the model & prior, the Bayesian solution is sound
- Model specification is always challenging
- Related question: where does the prior come from?
- Strategies:
 - full elicitation
 - partial elicitation & sets of priors
 - default/objective/noninformative priors
- Related to this: hierarchical and empirical Bayes

- Full elicitation is time consuming, unreliable,¹⁶ and rarely used
- One might pick a parametric family of priors and then elicit the hyperparameters from subject experts
- Knowing a few moments/quantiles would often be enough to determine the hyperparameters
- Even this limited type of elicitation is rarely done

¹⁶Evidence in the literature suggests experts can't formulate preferences in a way that's compatible with probabilities...

- This may be obvious, but *the prior matters*
- Recall the cancer screening example above
- Example:
 - prior for the mean of a Poisson distribution
 - suppose we know the 50th and 75th percentiles are 2 & 4, resp.
- Table¹⁷ shows the posterior mean for different observations x and three different parametric forms matching the constraints

Prior	x								
	1	2	3	4	5	10	15	20	50
Exp	1.49	2.23	2.97	3.71	4.46	8.17	11.88	15.60	37.87
log-N	1.48	2.11	2.81	3.56	4.35	8.66	13.24	17.95	47.02
log-Cau	1.56	2.09	2.63	3.25	3.98	8.87	14.07	19.18	49.40

¹⁷From Ghosh et al *Introduction to Bayesian Analysis*

- If one can't choose a single prior distribution, then it's possible to work with a set of plausible priors
- Called *robust* or *generalized Bayes*
- I think this is an appealing approach
- Some computational challenges, but doable in certain cases
- Sec. 5.6.4 in my book has some details about this, with an example of Walley's *imprecise Dirichlet model*
- I have a different (non-Bayesian) way to formulate this, which I can tell you about later

...[say] that the magnitude of a parameter is unknown, when none of the possible values need special attention. —Jeffreys

- Can we not just plug in a default prior Q and update?
- Should be wary about doing this, cf. Fraser's quote
- Intuitively, as Jeffreys says, such a default prior shouldn't favor any particular values of Θ
- Laplace's idea: just take a flat prior $q(\theta) \propto 1$
- Fisher (and others) showed that this generally doesn't work!
- Jeffreys came up with a way to dodge this criticism...

Definition.

If Θ is the uncertain parameter in a model $\{P_\theta : \theta \in \mathbb{T}\}$, and $I(\theta)$ is the Fisher information, then *Jeffreys prior* for Θ has PDF

$$q_{\text{JEF}}(\theta) \propto \{\det I(\theta)\}^{1/2}, \quad \theta \in \mathbb{T}$$

- If Θ is a location parameter, then $I(\theta)$ is constant, and so Jeffreys prior is flat
- Otherwise, Jeffreys prior isn't flat (in the usual sense¹⁸)
- The proof that this dodges Fisher's criticism is subtle; details are in Sec. 5.6.3 of my book
- Note: it's often the case that q_{JEF} is improper...

¹⁸There's a sense in which Jeffreys prior is flat relative to the geometric structure on $\mathbb{T}$ induced by the model

Exercise.

Let $X_1, \dots, X_n$ be iid $\text{Ber}(\theta)$. Find the Jeffreys prior and determine under what conditions the posterior is well-defined.

Exercise.

Let $X_1, \dots, X_n$ be iid $\text{Pois}(\theta)$. Find the Jeffreys prior and determine under what conditions the posterior is well-defined.

Exercise.

Let $X_1, \dots, X_n$ be iid $N(\theta, \sigma^2)$. Find the Jeffreys prior and show that the posterior is always well-defined.

- Even if a default prior is improper, there's often nothing mathematically wrong with using it
- If $\int L_n(\theta) q(\theta) d\theta < \infty$, then Q_n is well-defined
- This still fundamentally changes the interpretation of the posterior dist'n and the Bayes solution more generally
- The posterior isn't an update of prior beliefs anymore...

...any serious mathematician would surely ask how you could use [Bayes's theorem] with one premise missing by making up an ingredient and thinking that the conclusions of the [theorem] were still available. —Fraser

- With the use of default priors:
 - the connection to exchangeability is lost
 - the likelihood principle is violated
 - the posterior is no longer coherent
- Nevertheless, the Bayesian solution still generally “works” in the sense that point estimators, etc have good properties
- But then what does Bayes have to offer beyond the likelihood-based methods discussed above?

Satellite collision example.

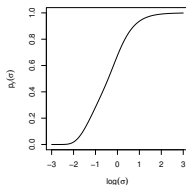
- An orbiting satellite could collide with another object
- Potentially major problem, collision should be avoided
- Decisions based on a *collision probability*
 - based on a standard default-prior Bayes solution
 - details in Balch, M., and Ferson, arXiv:1706.08565
- If collision probability is low, then satellite is judged safe; otherwise, some action is taken
- An unusual phenomenon has been observed in this literature:

noisier data makes non-collision probability large

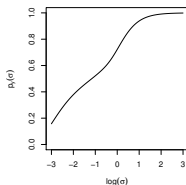
Satellite collision example, cont.

■ Simple illustration:

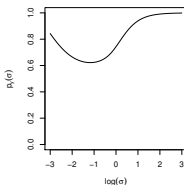
- $\|y\|$ denotes measured distance between satellite and object
- True distance ≤ 1 implies collision
- Measurement error variance is σ^2
- $p_y(\sigma) =$ probability of non-collision, given y
- When σ is large, $p_y(\sigma)$ is large, no matter what y is!



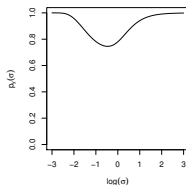
(a) $\|y\|^2 = 0.5$



(b) $\|y\|^2 = 0.9$



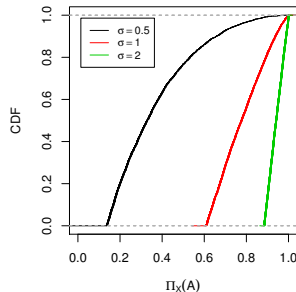
(c) $\|y\|^2 = 1.1$



(d) $\|y\|^2 = 1.5$

Satellite collision example, cont.

- Simplified version of the satellite example
 - let $A = \{\text{non-collision}\}$
 - then $Q_Y(A)$ as a random variable, with a CDF
 - plot CDF when data are generated under a collision course
- *False confidence*: $Q_Y(A)$ is almost always large!



- Just as with the likelihood, we confirm that Bayesian methods “work” is via asymptotics
- Roughly, Bayesian and likelihood methods have the same asymptotic properties
- No surprise, since Bayesian methods are likelihood-driven
- Shalizi: “frequentist license for Bayesian practice”
- Modern approach focuses on concentration of the (sequence of) posterior distributions

Definition.

The posterior is consistent at θ if, for all $\varepsilon > 0$,

$$\underbrace{Q_n(|\Theta - \theta| > \varepsilon)}_{\text{function of } X^n} \rightarrow 0 \quad \text{in } P_\theta\text{-probability}$$

Definition.

The posterior is r_n -consistent at θ if, for any $M_n \rightarrow \infty$,

$$\underbrace{Q_n(|\Theta - \theta| > M_n r_n)}_{\text{function of } X^n} \rightarrow 0 \quad \text{in } P_\theta\text{-probability}$$

Exercise.

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Pois}(\theta)$ and take Q as Jeffreys prior. Show that the posterior is $n^{-1/2}$ -consistent at every θ .

Exercise.

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Unif}(0, \theta)$ and take Q with PDF $q(\vartheta) = \vartheta^{-1}$. Show that the posterior is n^{-1} -consistent at every θ .

- Like with MLEs, consistency in examples is often easy, but a general consistency theorem is challenging
- The Kullback–Leibler divergence is defined as

$$K(f, g) = \int \log \frac{f(x)}{g(x)} f(x) dx$$

- KL-neighborhood: $\mathbb{K}_\kappa(\theta) = \{\vartheta : K(p_\theta, p_\vartheta) < \kappa\}$

Theorem.

If the prior satisfied the *KL-condition*, i.e.,

$$Q\{\mathbb{K}_\kappa(\theta)\} > 0 \quad \text{for all } \kappa > 0$$

then the posterior is consistent at θ

Exercise.

For a Poisson model, find $K(p_\theta, p_\theta)$ and check that it's ≥ 0

- Despite it's complicated-looking form, the KL-neighborhood is typically just an open set containing θ
- So, the KL-condition is met if prior has PDF q that's positive and continuous at θ
- The trouble is that we don't know the θ at which we need to satisfy the KL-condition
- To show it for “all” θ requires that the prior suitably covers $\mathbb{T}$
- Easy to do for low-dim problems, hard in high-dims

- Posterior consistency has an interesting consequence
- For any two priors that satisfy the conditions, the corresponding posterior dist'ns will eventually look the same
- This is called a *merging of opinions* phenomenon
- i.e., the effect of the prior washes out asymptotically
- Some see this as support the use of default priors...
- However, if the data is so informative that the choice of prior doesn't matter, then we shouldn't need to make a choice

Asymptotic properties, cont.

- General concentration rate theorems are available
- For the low-dim problems we're focusing on here, the posterior is generally $n^{-1/2}$ -consistent
- This is a consequence of a stronger result that parallels the asymptotic normality result for MLEs
- Roughly, Q_n will take on the shape of a normal distribution centered at the MLE with variance equal to the CR bound

Bernstein–von Mises theorem.

Under conditions similar to those of the MLE's asymptotic normality theorem, if Q has a positive & continuous PDF at θ , then

$$\int |q_n(\vartheta) - N(\vartheta \mid \hat{\theta}_n^{\text{MLE}}, \{nI(\hat{\theta}_n^{\text{MLE}})\}^{-1})| d\vartheta \rightarrow 0, \quad \text{in } P_\theta\text{-probability}$$

- Appearance of the CR bound implies “posterior efficiency”
- The metric uses here is *total variation distance*
- In particular, the quantiles of Q_n can be well-approximated by those of the limiting normal
- This implies that Q_n ’s credible sets will take the form

$$\hat{\theta}_n^{\text{MLE}} \pm \text{qnorm}(1 - \frac{\alpha}{2}) \times \{nl(\hat{\theta}_n^{\text{MLE}})\}^{-1/2}$$

- But we know that this is an asymptotic confidence set
- So, the BvM theorem implies that Bayesian credible sets are asymptotic confidence sets
- Shalizi: “Frequentist license for Bayesian practice”

- Bayesian framework is attractive
 - normative
 - fully probabilistic
 - reliable UQ (probability theory is self-consistent)
 - foundational benefits
 - ...
- Unfortunately, it's common in practice that we're lacking genuine prior information
- In these cases, most/all of what makes the Bayesian framework attractive disappears
- Default-prior Bayesian methods might still “work,” but that's not particularly impressive
- Fortunately, I think we can do better...

Exercise.

Let $(X_1, \dots, X_n \mid \Theta = \theta)$ be iid with *negative binomial* PMF

$$p_{\theta}(x) = \binom{x+r-1}{x} \theta^r (1-\theta)^x, \quad x = 0, 1, 2, \dots, \quad \theta \in (0, 1).$$

The number of successes r is known. If Θ has a $\text{Beta}(a, b)$ prior, then find the posterior distribution of Θ and the posterior mean

Exercise.

Let $(X_1, \dots, X_n \mid \Theta = \theta)$ are iid $\text{Unif}(0, \theta)$. A standard default prior has “PDF” $q(\theta) \propto \theta^{-1}$, for $\theta > 0$. Find a 95% credible interval and evaluate the coverage probability

Exercise — *Today's posterior is tomorrow's prior.*

For iid data, show that the posterior Q_{n+1} for Θ based on applying Bayes's theorem to the prior Q and the data set X^{n+1} as a batch is the same as that based on applying Bayes's theorem to the “prior” Q_n (the posterior based on data X^n) and the data point X_{n+1} alone.

Questions, etc?

- Decision theory
 - loss and risk functions
 - admissibility
 - minimizing global measures of risk
 - Bayes rules
 - minimax rules
 - minimizing risk under constraints
 - best unbiased estimation
 - most powerful size- α tests
 - high-dimensional cases

- So far, we've focused mainly on two frameworks
 - likelihood
 - Bayesian
- The theory investigates the *frequentist properties* of the various methods, e.g., consistency
- Consistent with the inductive logic at the start
- Here we go all-in on frequentism, i.e., we aim to find the “best” methods in some specific sense
- Different from our previous discussions:
 - we're not starting from a given framework that we like and asking about its properties
 - goal is to find procedures, whatever they are, that optimize these frequentist properties

- Classical approach is to frame the statistical inference problem in terms of decision-making
- e.g., point estimation is deciding on a (data-driven) guess of the unknown value of the parameter
- To formalize this we need
 - a set of possible actions $\mathbb{A}$
 - a loss function $(\theta, a) \mapsto L(\theta, a)$ describes the loss incurred by taking action $a \in \mathbb{A}$ when the world is in state $\theta \in \mathbb{T}$
- For example:
 - $\mathbb{A} = \phi(\mathbb{T})$ and $L(\theta, a) = \{a - \phi(\theta)\}^2$
 - $\mathbb{A} = \{0, 1\}$ and $L(\theta, a) = |1(\theta \in H_a) - a|$

- Of course, since we have data X^n , we'd like to use this info to help choose an action
- A *decision rule* is a function¹⁹ $\delta : \mathbb{X}^n \rightarrow \mathbb{A}$
- Interpretation: $\delta(X^n)$ is a data-driven decision
- Notation is new to us, but the idea isn't:
 - estimators are decision rules
 - so are our tests & confidence sets
 - p-value functions & posterior dist'ns are too... I think
- Quality of a decision rule is measured by $L(\theta, \delta(X^n))$
- Not directly useful, however...
 - it's noisy (a random variable)
 - we don't know which θ to focus on

¹⁹Technically, δ could return a probability dist'n on $\mathbb{A}$, randomization...

Definition.

For a decision problem with loss L , the risk function associated with decision rule δ is

$$R(\theta, \delta) = \mathbb{E}_{\theta}\{L(\theta, \delta(X^n))\}, \quad \theta \in \mathbb{T}$$

- Risk removes the noise from individual data X^n
- Measures on-average quality of δ if the world was in state θ
- Goal: find δ to minimize $R(\cdot, \delta)$ in some sense
- Non-trivial because $R(\cdot, \delta)$ is a *function* for each δ , i.e., how can we minimize a function-valued function?

Definition.

A decision rule δ is *inadmissible* if there exists δ' such that

$$R(\theta, \delta') \leq R(\theta, \delta) \quad \text{for all } \theta, \text{ strict } < \text{ for some } \theta$$

δ is *admissible* if it's not inadmissible

- Before we try to find the “best” δ , let's focus our attention on the smaller collection that are *admissible*
- To a pure frequentist, there's really no good reason to consider using an inadmissible rule
- So it's important to our overall goal that we can characterize those rules that are admissible

- The following result connects admissibility to sufficient statistics, which is really cool
- Any decision rule that's not a function of the sufficient statistic can be improved upon via conditioning

Rao–Blackwell theorem.

Let δ be a decision rule and T a sufficient statistic for $\{P_\theta : \theta \in \mathbb{T}\}$. If $a \mapsto L(\theta, a)$ is convex for all θ , then the decision rule

$$\delta^{\text{RB}}(X^n) = E\{\delta(X^n) \mid T\} \quad \leftarrow \text{only depends on } T!$$

satisfies $R(\theta, \delta^{\text{RB}}) \leq R(\theta, \delta)$ for all θ

Exercise — True or False?

The Rao–Blackwell theorem implies that if an estimator depends on only a minimal sufficient statistic, then it is admissible.

Exercise.

$X^n \stackrel{\text{iid}}{\sim} \text{Pois}(\theta)$; goal is to estimate $\phi(\theta)$ under squared error loss.

- 1 Let $\phi(\theta) = e^{-\theta}$. Use the Rao–Blackwell to get an estimator that dominates the naive $\delta(X^n) = 1(X_1 = 0)$
- 2 Let $\phi(\theta) = \theta^2$. Use the Rao–Blackwell to get an estimator that dominates the naive $\delta(X^n) = X_1 X_2$

Exercise.

When does $R(\theta, \delta^{\text{RB}}) = R(\theta, \delta)$ for all θ ?

- Admissibility is important but, on its own, it's rather weak
- Lots of bad decision rules are admissible, e.g., an estimator $\delta(X^n) \equiv 7$ that ignores the data is admissible
- What's the issue?
 - most decision rules have risk functions that intersect
 - too ambitious to rank δ s by risk uniformly over all θ
- This observation suggests alternative approaches:
 - compare risks based on a (global) summary
 - impose constraints on the decision rules

- Two operations that we commonly apply to functions are
 - integration
 - optimization
- Loosely speaking, these offer global summaries of the function
- In our context, this amounts to
 - average risk, $r(Q, \delta) = \int R(\theta, \delta) Q(d\theta)$
 - worst-case risk, $r(\mathbb{T}, \delta) = \sup_{\theta \in \mathbb{T}} R(\theta, \delta)$
- We'll talk about both of these in turn

Definition.

Given a “prior” Q , the average/Bayes risk is

$$r(Q, \delta) = \int R(\theta, \delta) Q(d\theta)$$

The corresponding Bayes rule, δ_Q , minimizes $\delta \mapsto r(Q, \delta)$

- If Q is a proper assessment of one's subjective prior beliefs about θ , then this is quite natural
- Aligns with the von Neumann–Morganstern theory
- Different perspective: we can adjust Q to ensure that δ_Q has the properties that we want...
- How to find the Bayes rule?

Theorem.

Sidestepping technicalities, the Bayes rule can be expressed as

$$\delta_Q(x^n) = \arg \min_a \underbrace{\int L(\theta, a) Q_{x^n}(d\theta)}_{\text{posterior risk, given } x^n}$$

- This pointwise-in- x^n definition is much easier to evaluate than directly minimizing $r(Q, \delta)$
- This explains some previous terminology:
 - squared-error loss, Bayes estimator is the posterior mean
 - zero-one loss, Bayes test rejects when $Q_{x^n}(\cdot) < \frac{1}{2}$

Proof of the claim about Bayes tests under zero-one loss.

Two (exhaustive) hypotheses H_0 and $H_1 = \mathbb{T} \setminus H_0$, with action space $\mathbb{A} = \{0, 1\}$. Posterior risk of the two actions

$$\int L(\theta, 0) Q_{x^n}(d\theta) = Q_{x^n}(H_1)$$

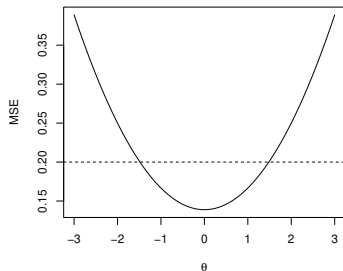
$$\int L(\theta, 1) Q_{x^n}(d\theta) = Q_{x^n}(H_0)$$

We'd choose action $a = 1$, i.e., reject H_0 , iff

$$Q_{x^n}(H_0) < Q_{x^n}(H_1) \iff Q_{x^n}(H_0) < \frac{1}{2}.$$

Average/Bayes risk, cont.

- Interesting observation (to be made a bit more formal):
 - Bayes rules perform well at θ values consistent with Q
 - roughly, Bayes rules can't be beat near prior center
 - suggests that Bayes rules are admissible
- Example: $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$
 - sample mean has constant risk
 - if $Q = N(0, 1)$, then Bayes rule is better near origin



Theorem.

Under some mild technical conditions, if the prior Q is proper (total mass 1) with PDF q and if $\theta \mapsto R(\theta, \delta)$ is continuous for each δ , then the Bayes rule δ_Q is admissible

- The conditions are quite mild and apply in a wide range of examples, e.g., exponential families
- Various relaxations/extensions are possible:
 - handling improper (default) priors
 - handling sequences of (improper) priors
- This is generally how admissibility is proved:
 - show that the given decision rule is Bayes wrt ...
 - apply a version of the Bayes-rules-are-admissible theorem

- These are remarkable results!
- Roughly, this means that even the pure frequentists ought to be at least approximately Bayesian
- *Complete-class theorems*: under certain conditions, the only admissible decision rules are (limits of) Bayes rules
- Recall: admissibility is relatively weak property on its own
- To accept a Bayes rule because its a Bayes rule, we need genuine prior info to point us in the right direction

Definition.

The *minimax rule* δ^{MM} minimizes the maximum risk, i.e.,

$$\sup_{\theta} R(\theta, \delta^{\text{MM}}) = \inf_{\delta} \sup_{\theta} R(\theta, \delta)$$

- Minimax rules are conservative, focus on the worst-case
- In game theory, minimax rules make sense when play is against an adversarial opponent
- Is “Nature” an adversarial opponent...?

- Defining a minimax rule is one thing, but how to find it?
- Interestingly, there's another connection to Bayes
- Roughly: very peculiar Bayes rules are minimax
- “Peculiar” in the sense that Bayes risk = max risk

Theorem.

Let Q be a proper prior and δ_Q the corresponding Bayes rule. If $r(Q, \delta_Q) = \sup_{\theta} R(\theta, \delta_Q)$, then δ_Q is minimax

Exercise.

Prove the above theorem

Exercise.

Prove the following corollary: *If a Bayes rule has constant risk, then it's minimax*

Exercise.

Show that if a decision rule is admissible and has constant risk, then it's minimax

Example — Binomial minimax estimator.

If $Q = \text{Beta}(a, b)$, then $\delta_Q(x) = (a + x)/(a + b + n)$. Choose (a, b) such that risk is constant? The risk function is quadratic

$$R(\theta, \delta_Q) = A\theta^2 + B\theta + C,$$

where

$$A = \frac{(a + b)^2 - n}{(a + b + n)^2}, \quad B = \frac{n - 2a(a + b)}{(a + b + n)^2}, \quad C = \frac{a^2}{(a + b + n)^2}$$

Can be shown that $A = B = 0$ iff $a = b = n^{1/2}/2$, so

$$\delta^{\text{MM}}(x) = \frac{x + n^{1/2}/2}{n + n^{1/2}}$$

- For the finite-dim problems we've been focusing on here, minimax rules are sort of extreme
- But it's more relevant in high-dim problems, so it's worth trying to get whatever insight we can out of this
- Key idea: minimax rules are Bayes wrt certain priors
- Those “peculiar” priors are called *least favorable*, i.e.,

$$r(Q^{\text{LF}}, \delta_{Q^{\text{LF}}}) \geq r(Q, \delta_Q) \quad \text{for all } Q$$

- Then the general strategy for finding minimax risks, and rules that attain them, is constructing least favorable priors

Minimizing risk under constraints

- As an alternative to minimizing a summary of the risk function, let's impose constraints on the candidate δ 's
- There's no guarantee that a given constraint would admit a decision rule that's optimal uniformly over θ
- The two kinds of constraints I'll focus on are
 - estimators that are unbiased
 - tests with a fixed size/Type I error probability
- Under these constraints, optimality results are possible
- Intuition (in the estimation context):
 - recall that constant estimators are admissible
 - enforcing an unbiasedness eliminates constant estimators

Minimizing risk under constraints, cont.

- Consider estimation of $\phi(\theta)$ under squared error loss
- Then the risk is sometimes called the *mean square error*
- Bias–variance decomposition:

$$R(\theta, \delta) = V_{\theta}\{\delta(X^n)\} + \underbrace{[E_{\theta}\{\delta(X^n)\} - \phi(\theta)]^2}_{\text{bias}}$$

- If δ is unbiased, then the second term vanishes; so the focus turns exclusively to variance

Definition.

The uniformly minimum variance unbiased estimator satisfies

$$V_{\theta}(\delta^{\text{UMVUE}}) \leq V_{\theta}(\delta) \quad \text{for all unbiased } \delta \text{ and all } \theta$$

- While Bayes rules were powerful in the global risk minimization case, they're irrelevant here
- Fact:²⁰ proper-prior Bayes estimators are biased
 - intuition is that a Bayes rule shrinks towards prior center
 - this is basically why Bayes rules are admissible

Exercise.

Check that the Bayes estimator of the binomial success probability, under a conjugate $\text{Beta}(a, b)$ prior, is biased for all (a, b)

²⁰This isn't exactly true, there are weird/impractical priors that admit unbiased Bayes estimators

- Existence of a UMVUE remains to be addressed, but we can first prove uniqueness given existence
- This already is of some value:
 - in certain cases, we have estimators that attain the CR bound
 - so there's no other estimator with smaller variance
 - uniqueness means that it's *the* UMVUE

Theorem.

If it exists, the UMVUE is unique

- Towards existence, suppose we have a candidate UMVUE $\hat{\phi}_n$
- Then any other unbiased estimator can be expressed as

$$\hat{\phi}_n^s = \hat{\phi}_n + s \hat{0}_n$$

where $s \in \mathbb{R}$ and $\hat{0}_n$ is an *unbiased estimator of 0*

- This characterization is important because we get a concrete link between the variance of $\hat{\phi}_n$ and any other $\hat{\phi}_n^s$

$$V_{\theta}(\hat{\phi}_n^s) = V_{\theta}(\hat{\phi}_n) + s^2 V_{\theta}(\hat{0}_n) + 2s C_{\theta}(\hat{\phi}_n, \hat{0}_n)$$

- In particular, the covariance $C_{\theta}(\hat{\phi}_n, \hat{0}_n)$ is relevant...

Minimizing risk under constraints, cont.

- Specifically, $\hat{\phi}_n$ is the UMVUE iff its uncorrelated with every unbiased estimator of 0, i.e., iff

$$C_{\theta}(\hat{\phi}_n, \hat{0}_n) = 0 \quad \text{for all } \hat{0}_n \text{ and all } \theta$$

- In general, this is hard to check...
- But suppose there's a *complete & sufficient statistic* T
 - by RB, focus on estimators that only depend on T
 - completeness implies there's only one unbiased estimator of 0
 - all estimators are uncorrelated with constants
- Hence, at most one unbiased estimator that's a function of T

Lehmann–Scheffe theorem.

If T is a complete sufficient statistic and $\phi(\theta)$ is unbiasedly estimable, then there exists a unique UMVUE, a function of T

Minimizing risk under constraints, cont.

- If we have a complete sufficient T and can find an unbiased estimator that's a function of T , then it's the UMVUE
- e.g., in the $\text{Pois}(\theta)$ case
 - both $\bar{X}$ and S^2 are unbiased
 - the former is complete & sufficient, hence UMVUE
 - we find that $\bar{X}$ is better than S^2 without calculating $V_\theta(S^2)$!
- When T is complete & sufficient, RB returns the UMVUE
- We can also get optimality conclusions in cases where the CR bound doesn't apply, e.g., in $\text{Unif}(0, \theta)$
- But even existence of complete & sufficient stats is restrictive

Exercise.

For iid $\text{Pois}(\theta)$ data X^n , we want to estimate $\phi(\theta) = \mathbb{E}_\theta\{g(X_1)\}$ for a given scalar function g .

- 1 Show that $\delta(X^n) = n^{-1} \sum_{i=1}^n g(X_i)$ is an unbiased estimator of $\phi(\theta)$
- 2 Suppose g is non-linear. Use Rao–Blackwell to prove that $\delta(X^n)$ above is inadmissible
- 3 If g is linear, then prove that $\delta(X^n)$ is the UMVUE

Exercise.

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(\theta)$. The goal is estimation of $\phi(\theta) = \theta^2$.

- 1 Show that $\hat{\phi}_n = (1 + \frac{1}{n})^{-1} \bar{X}^2$ is unbiased
- 2 Show that $\hat{\phi}_n$ is the UMVUE

Minimizing risk under constraints, cont.

- Consider testing H_0 versus H_1
- A test is a function $\delta : \mathbb{X}^n \rightarrow \mathbb{A} = \{0, 1\}$
- Two relevant features
 - $\text{size}(\delta) = \sup_{\theta \in H_0} P_{\theta}(\delta = 1)$
 - $\text{pow}(\theta; \delta) = P_{\theta}(\delta = 1)$, $\theta \in H_1$
- Ideally, we want size small and power large
 - but they're tied together
 - e.g., $\text{size} \rightarrow 0$ implies $\text{pow} \rightarrow 0$
 - no test that uniformly minimizes size & maximizes power
- Now-standard idea is to fix a size and maximize the power relative to that constraint

Example.

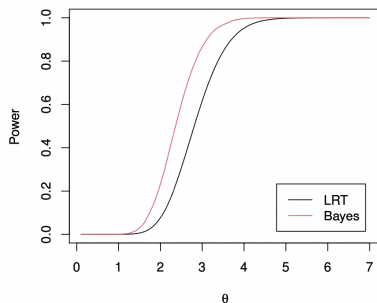


Figure 6.2: Power functions for the likelihood ratio and Bayes tests of $H_0 : \theta \leq \theta_0$, where $n = 20$, $\theta_0 = 2$, and the prior hyperparameters are $a = b = 3$.

Minimizing risk under constraints, cont.

Definition.

Let $\Delta_\alpha(H_0, H_1)$ denote the set of all tests δ with size equal to α . The uniformly most powerful (UMP) test $\delta_\alpha^{\text{UMP}}$ satisfies

$$\text{pow}(\theta; \delta_\alpha^*) \geq \text{pow}(\theta; \delta_\alpha) \quad \text{all } \theta \in H_1 \text{ and } \delta_\alpha \in \Delta_\alpha(H_0, H_1)$$

Neyman–Pearson lemma.

Let $H_0 = \{\theta_0\}$ and $H_1 = \{\theta_1\}$ and, for simplicity, suppose that data are continuous. Then the UMP size- α test is given by

$$\delta_\alpha^{\text{UMP}}(X^n) = 1 \left\{ \frac{p_{\theta_1}(X^n)}{p_{\theta_0}(X^n)} > c_\alpha \right\}$$

where $c_\alpha = c_\alpha(\theta_0)$ is chosen to satisfy

$$P_{\theta_0} \left\{ \frac{p_{\theta_1}(X^n)}{p_{\theta_0}(X^n)} > c_\alpha \right\} = \alpha$$

- Discrete data problems are also covered by the theorem, but there's some complications — *randomized tests*
- The simple-vs-simple testing problem isn't common in practice
- But Neyman & Pearson's result was groundbreaking, caused a paradigm shift in statistical thinking
- Their result can be extended a bit, but we don't have much beyond this in terms of optimality

Minimizing risk under constraints, cont.

Exercise.

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Exp}(1/\theta)$ with unknown rate parameter $\theta > 0$. Find the UMP test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$, where $\theta_1 > \theta_0$.

Exercise.

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$. Find the UMP test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$, where $\theta_1 > \theta_0$.

Exercise.

Let $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Beta}(\theta, 1)$ with unknown $\theta > 0$. The marginal PDF is

$$p_\theta(x) = \theta x^{\theta-1}, \quad x \in [0, 1]$$

Find the UMP test of $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$, where $\theta_1 > \theta_0$.

Minimizing risk under constraints, cont.

- In the examples above, the UMP test doesn't depend on the alternative θ_1 , only the direction $\theta_1 > \theta_0$ or $\theta_1 < \theta_0$
- This suggests the possibility of extending the scope of the most-powerful conclusion to a wider alternative
- The special structure that allowed the dependence on θ_1 to drop out is monotonicity in the likelihood ratio

Definition.

A family $\{g_\theta : \theta \in \mathbb{R}\}$ of univariate PMFs/PDFs has the *monotone likelihood ratio* property if, for each $\theta_1 > \theta_0$,

$$t \mapsto \frac{g_{\theta_1}(t)}{g_{\theta_0}(t)} \text{ is monotone (non-increasing or non-decreasing)}$$

Karlin–Rubin theorem.

Consider testing $H_0 = \{\theta_0\}$ versus $H_1 = (\theta_0, \infty)$. If T is a sufficient statistic with the monotone likelihood ratio property, then the UMP size- α test is of the form “reject H_0 iff $T > t_0$ ”

- Exponential families admit a sufficient statistic whose distribution satisfies the monotone likelihood ratio property
- So there are lots of cases where the Neyman–Pearson test is UMP for composite alternatives
- My statement above has a simple null, but the same monotonicity allows extension to composite null

Minimizing risk under constraints, cont.

- There's no UMP test for " $\theta = \theta_0$ " vs " $\theta \neq \theta_0$ "
- Need an additional "unbiasedness" constraint...
- Plot shows the power function of the two one-sided UMP tests, along with the UMP-unbiased test's power function

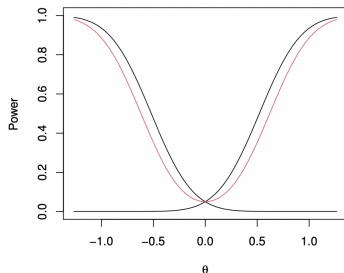


Figure 6.4: Plots of the power functions for the three tests considered in the normal example, i.e., the two UMP tests (black) for one-sided alternatives and the natural test (red) for the two-sided alternative. Here I'm using $n = 10$, $\theta_0 = 0$, $\sigma = 1$, and $\alpha = 0.05$.

- I've implicitly focused on low-dim cases so far
- There are some important differences in high-dim cases, and my plan is to briefly describe these here
- First, by “high-dim” what I mean is that *there are as many parameters as data points*
- Challenge: the “info increases with n ” feature that drives, e.g., consistency, in low-dim cases no longer holds
- So, more care is needed in general

- My favorite “many-normal-means” problem
 - $X_i \stackrel{\text{ind}}{\sim} N(\theta_i, 1)$, for $i = 1, \dots, n$
 - high-dim because each data point has it's own parameter
 - difficult enough to showcase challenges
- Let $\theta = (\theta_1, \dots, \theta_n)$ be the quantity of interest
- Squared-error loss, $L(\theta, a) = \|a - \theta\|^2 = \sum_{i=1}^n (a_i - \theta_i)^2$
- $\delta(X) = X$ is the MLE, it's unbiased, etc
- The following striking result makes clear that there's something different about high-dim cases

Stein's inadmissibility theorem.

For $n \geq 3$, the MLE $\delta(X) = X$ is inadmissible

- Stein's proof constructs an estimator that dominates the MLE
- This is now called the James–Stein estimator

$$\delta^{\text{JS}}(X) = \left(1 - \frac{n-2}{\|X\|^2}\right) X$$

- δ^{JS} has some Bayesian connections...
- Unexpected/paradoxical consequences of the additive loss:
 - despite there being no connection between θ_i and θ_j , information in data is shared across estimands
 - there's a force that wants to shrink data towards the origin
- Stein's result²¹ was the genesis of the modern focus on *shrinkage estimation* and regularization more generally

²¹Ironically, δ^{JS} is inadmissible too, but the result was still super-impactful...

- The scientific problem often suggests that the high-dim quantity of interest has a latent lower-dim structure
- The structure that I'll focus on here is *sparsity*
- In words, sparsity = “most of the θ_i 's are small/zero”
- Mathematically, sparsity means $\theta \in \mathbb{T}_n(s_n)$ where
 - $s_n = o(n)$
 - $\mathbb{T}_n(s) = \{\theta \in \mathbb{R}^n : \|\theta\|_0 \leq s\}$
- The MLE, which isn't sparse, turns out to be pretty bad:

$$R(\theta, \delta^{\text{MLE}}) = \sum_{i=1}^n \mathbb{E}_{\theta_i} (X_i - \theta_i)^2 = n$$

- For comparison, suppose an oracle told us which of the θ_i 's are non-zero, i.e., $S = \{i : \theta_i \neq 0\}$
- Then the modified MLE is

$$\delta^{\text{OR}}(X)_i = X_i 1(i \in S), \quad i = 1, \dots, n$$

and its risk is

$$R(\theta, \delta^{\text{OR}}) = \sum_{i \in S} \mathbb{E}_{\theta_i} (X_i - \theta_i)^2 = |S| \leq s_n \ll n$$

- δ^{OR} is much better than δ^{MLE} , but is unrealistic
- Our target is a risk that grows at rate between s_n and n

Theorem.

The minimax risk for estimation of a sparse normal mean is

$$\inf_{\delta} \sup_{\theta \in \mathbb{T}_n(s_n)} R(\theta, \delta) \sim O(s_n \log(n/s_n)), \quad n \rightarrow \infty$$

- The proof is based on constructing a least favorable prior distribution for θ as discussed earlier
- An *asymptotically minimax rule* is one whose maximum risk grows at the same rate as above
- There are lots of asymptotically minimax estimators
- Consider a *hard thresholding estimator*

$$\delta_t^{\text{HT}}(X)_i = X_i \mathbf{1}(|X_i| > t), \quad i = 1, \dots, n$$

Theorem.

Relative to $\mathbb{T}_n(s_n)$, the hard thresholding estimator with $t = \{2 \log(n/s_n)\}^{1/2}$ is asymptotically minimax

- It's not too difficult to achieve asymptotic minimaxity in this context; other applications are more difficult
- A shortcoming is that the above requires s_n be known
 - an *adaptively asymptotically minimax* estimator would achieve the minimax rate without knowledge of s_n
 - using $t = (2 \log n)^{1/2}$ works well, but there are better options
- Lots of modern/recent theoretical work in the literature involves similar analyses in more complex problems

- Decision theory aims to find the “best” procedures, in terms of pure frequentist considerations
- Ends up closely aligning things we already discussed
- Unfortunately, we can’t find universally best procedures
- Instead, we can get optimality relative to
 - global measures of risk
 - constrained decision rules
- Classical theory in simple problems
- Extension to high-dim problems, a paradigm shift in statistical inference — *regularization* (e.g., lasso)

Questions, etc?

- Higher-level coverage of two topics...
- bridging the gap between frequentist and Bayes
 - a bit about Fisher's fiducial argument
 - my *inferential model* framework
- inference on quantities not determined by a statistical model
 - generalization of the likelihood-based framework
 - generalization of the Bayesian framework

How can a discipline, central to science and to critical thinking, have two methodologies, two logics, two approaches that frequently give substantially different answers to the same problems? —Fraser

Unlike most philosophical arguments, this one has important practical consequences. The two philosophies represent competing visions of how science progresses and how mathematical thinking assists in that progress. —Efron

- As we know, Fisher made many contributions to statistics
 - likelihood & MLEs
 - sufficiency
 - efficiency
 - ancillary statistics & conditioning
 - ...
- What many people don't know is that these efforts were part of Fisher's push for an alternative to Bayesian inference
 - he didn't believe subjective priors should be part of science
 - but he still wanted "probabilistic inference"

[Fisher's fiducial inference is] a bold attempt to make the Bayesian omelet without breaking the Bayesian egg. —Savage

- Fisher's idea was to construct a sort of “posterior dist'n” for Θ using the model and data only
- He called his proposal the *fiducial argument* and the output of his proposed approach *fiducial probabilities*
- Fiducial = “based on trust or faith”
- To Fisher, the “trust” element of his proposal is related to the inductive logic from earlier
 - fiducial probability of a false hypothesis might be high
 - but the answer being wrong is a small-probability event

Fisher's (first) fiducial argument.

- X and Θ are scalars, model w/ continuous CDF $x \mapsto P_\theta(x)$
- For any $p \in [0, 1]$, there exists $x_p(\theta)$ and $\theta_p(x)$ with

$$p = F_\theta(x_p(\theta)) = F_{\theta_p(x)}(x)$$

- Certain “monotonicity” in the model leads to
 - $x \leq x_p(\theta) \iff \theta \geq \theta_p(x)$ for all (p, x, θ)
 - $p = P_\Theta\{X \leq x_p(\Theta)\} = P_\Theta\{\Theta \geq \theta_p(X)\}$
- Fisher *defines* the fiducial dist'n for Θ , given $X = x$, via

$$“P\{\Theta \geq \theta_p(x)\}” = p, \quad p \in [0, 1]$$

- Corresponds to a PDF $\theta \mapsto |\partial P_\theta(x)/\partial \theta|$

Exercise.

For a location parameter model, with $p_\theta(x) = p(x - \theta)$, show that the fiducial distribution has PDF $\theta \mapsto p(x - \theta)$, which agrees with the Bayes solution using a flat prior

Exercise.

For a scale parameter model, with $p_\theta(x) = \frac{1}{\theta} p(\frac{x}{\theta})$, show that the fiducial distribution has PDF $\theta \mapsto \frac{x}{\theta} p_\theta(x)$, which agrees with the Bayes solution using default prior $q(\theta) \propto \frac{1}{\theta}$

- Fisher's argument requires $\dim(\text{data}) = \dim(\text{parameter})$
 - so he needed to justify data reduction
 - motivated sufficiency, efficiency, conditioning, etc
- Connections:
 - fiducial almost indistinguishable from “confidence limits”²²
 - fiducial solution often is similar to, and sometimes the same as default-prior Bayes solutions
- But Fisher wasn't thinking about confidence limits
- He wanted data-dependent probabilities that are reliable in our inductive logic sense

²²But Neyman's first formulation of confidence limits was based on his interpretation of Fisher's fiducial argument...

- Fisher's proposal was quite mysterious and motivated many other researchers to understand
- This had major impacts on the discipline!
- But Fisher's arguments couldn't stand up to all the scrutiny
- For example, a major concern was a non-uniqueness that emerged in certain multiparameter problems
- Eventually became "Fisher's biggest blunder" (Efron)
- But that's not the end of the story...

There is still more to be learned from Fisher, even when he seems clearly wrong, than from any other contributor to statistical thinking. —Dawid

Fisher's attempt to steer a path between the Scylla of unconditional behaviorist methods which disavow any attempt at "inference" and the Charybdis of subjectivism in science was founded on important concerns, and his personal failure to arrive at a satisfactory solution to the problem means only that the problem remains unsolved, not that it does not exist. —Zabell

Perhaps the most important unresolved problem in statistical inference is the use of Bayes theorem in the absence of prior information. —Efron

- Lots of other ideas have been put forward
 - default-prior Bayes
 - generalized fiducial inference
 - confidence distributions
 - Dempster–Shafer theory
 - ...
- None of these alternatives are widely accepted
- I think it's because they generally lack a connection to the inductive logic that was key to Fisher (and to science)

- The trouble is that Fisher was too ambitious
- The model alone isn't informative enough to reliably support *exact probability statements*
- He knew this:
 - *[A p-value]... does not justify any exact probability statement*
 - *no exact probability statements can be based on [conf. sets]*
- That is, the things that are “reliable” — namely p-values and confidence limits — aren't exact probabilities
- But the theory needed to make this rigorous was lacking and Fisher ultimately fell short
- But it's still not the end of the story...

- Relatively new insights...²³
- I claim that the following is problematic:
 - for some false hypotheses $H \not\in \Theta$
 - the rv $X \mapsto Q_X(H)$ tends to be large under P_Θ
- *False confidence* is a particular form of unreliability that all probabilistic UQ suffers from

False confidence theorem.

Let Q_X be *any* data-dependent probability for Θ . For any $\alpha \in (0, 1)$, there exists hypotheses $H \subset \mathbb{T}$ such that

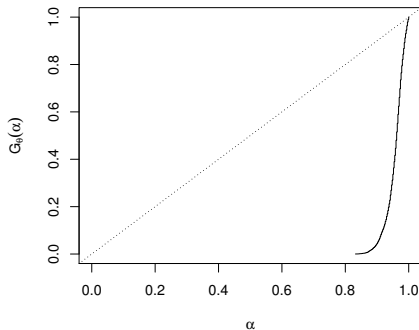
$$\underbrace{H \not\in \Theta}_{H \text{ is false}} \quad \text{and} \quad P_\Theta\{\underbrace{Q_X(H) > 1 - \alpha}_{\text{confidence in } H}\} > \alpha.$$

²³Balch, M., & Ferson 2019, *Proc Roy Soc A*, arXiv:1706.08565

- Example: Fieller–Creasy problem
- Model: $X \sim P_{\Theta} = N_2(\Theta, I)$, with $\Theta = (\Theta_1, \Theta_2)$
- Standard posterior: $Q_x = N_2(x, I)$.
- Goal is to quantify uncertainty about the ratio $\Phi = \Theta_1/\Theta_2$
- Let $H = \{\theta : \theta_1/\theta_2 \leq 9\}$
- Suppose $\Theta = (1, 0.1)$, so that $\Phi = 10$ and H is false
- We'd expect $Q_X(H)$ to be small
- Let's see...

False confidence, cont.

- Define the CDF: $G_\theta(\alpha) = P_\theta\{Q_X(H) \leq \alpha\}$.
- Plot uses $\theta = (1, 0.1)$, so $H = \{\theta_1/\theta_2 \leq 9\}$ is false
- $Q_X(H)$ actually tends to be *large* — false confidence!



- Original result in Balch et al only establishes *existence* of hypotheses afflicted by false confidence
- Could all the afflicted hypotheses be trivial/impractical?
 - my experience said *No*
 - but I wasn't able to make any progress on characterizing the problematic hypotheses
- Very recent paper (arXiv:2404.16228) gets me closer...

Which statistical hypotheses are afflicted with false confidence?

- Data + model can't reliably support precise probabilities
- Is it really necessary that probabilities be *precise*?
- Data-dependent *imprecise probabilities* for (reliable) UQ...

Definition — *inferential model* (IM).

- given data x , model, etc
- to every $H \subseteq \mathbb{T}$, assign a pair of numbers

$$\underline{\Pi}_x(H) \quad \text{and} \quad \overline{\Pi}_x(H)$$

- x -dependent measures of support for and plausibility of H

- *Possibility theory*²⁴ is the simplest imprecise probability
- Similar to probability theory
 - determined by a “density function”
 - but with a different calculus — optimization vs integration
- Specifically:
 - possibility contour $\theta \mapsto \pi_x(\theta)$ with $\sup_{\theta} \pi_x(\theta) = 1$
 - define $\overline{\Pi}_x(H) = \sup_{\theta \in H} \pi_x(\theta)$
 - $\underline{\Pi}_x(H) = 1 - \overline{\Pi}_x(H^c)$
- You’ve already seen all this — p-values!
- i.e., our likelihood-based p-values define a *possibilistic IM*

²⁴Zadeh 1978, Dubois & Prade 1984, etc...

IM validity theorem.

The p-value-based possibilistic IM is *valid*, i.e.,

$$\sup_{\theta \in H} P_{\theta}\{\pi_X(\theta) \leq \alpha\} \leq \alpha, \quad \alpha \in [0, 1]$$

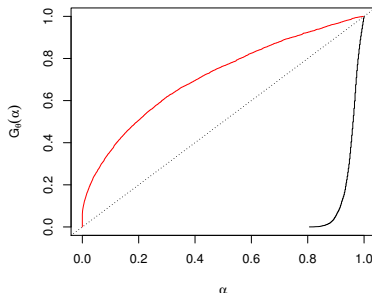
This implies that, for any $H \subseteq \mathbb{T}$,

$$\sup_{\theta \in H} P_{\theta}\{\bar{\Pi}_X(H) \leq \alpha\} \leq \alpha \quad \text{and} \quad \sup_{\theta \notin H} P_{\theta}\{\underline{\Pi}_X(H) > 1 - \alpha\} \leq \alpha$$

- Implications:
 - no false confidence
 - tests & confidence sets have error control guarantees²⁵
 - ...
- Need not require anything that we didn't already know

²⁵There's also a sort of "converse" (M., arXiv:2112.10904)

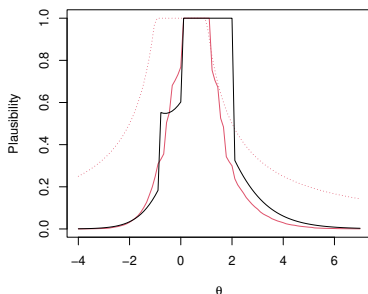
- Same Fieller–Creasy simulation as before
- Compare CDFs of $Q_X(H)$ and $\underline{\Pi}_X(H)$ ²⁶
- H is false, validity implies CDF is above diagonal
- False confidence vs. **no false confidence**



²⁶Based on contour $\pi_x(\theta) = 1 - \text{pchisq}(\|x - \theta\|^2, 2)$

Example: two binomials

- Clinical trial:
 - association between treatment & control?
 - let Θ denote the log odds ratio²⁷
- Data $(n_1, y_1) = (43, 1)$ and $(n_2, y_2) = (39, 2)$
- Focus on the black curve only



²⁷Nuisance parameters eliminated via conditioning (M., arXiv:2309.13454)

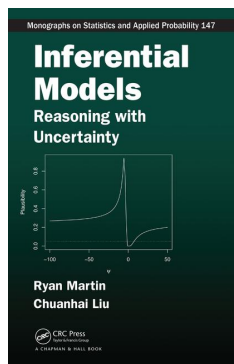
Example: Behrens–Fisher

- Difference of normal means, general variances²⁸
- Validity is guaranteed, simulation to check efficiency:
 - difficult unbalanced case, $(n_1, n_2) = (2, 20)$
 - compare coverage prob of 90% confidence intervals

Method	Coverage Prob
Hsu–Scheffe	0.9738
Jeffreys	0.9296
Ghosh & Kim	0.7873
Welch	0.8362
1st order	0.7399
Fraser et al	0.8617
IM	0.9082

²⁸Nuisance parameters eliminated via profiling (M., arXiv:2309.13454)

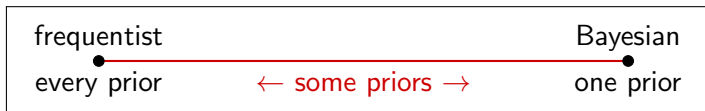
- First IM book →
- New n -part series of working papers:
*Valid and efficient imprecise-probabilistic
inference with partial priors*
- $n = 3$ parts so far
 - 1 arXiv:2203.06703
 - 2 arXiv:2211.14567
 - 3 arXiv:2309.13454
- Working towards a new book...



- Formal decision theory (arXiv:2112.13247)
- Machine learning problems:

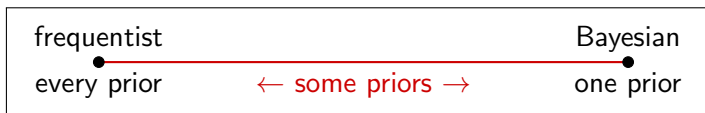


- Probabilistic reasoning is human nature
- Statisticians have a strong desire to quantify uncertainty about unknowns using probability
- Desire is so strong that they're willing to break the rules (improper priors) just to get it
- Fisher claimed it could be done without priors & Bayes theorem, via a *fiducial argument*
- Strong idea, but the theory doesn't hold up
- To get things to work, one must use *imprecise probability*
- Only a slight departure from familiar things...



- But IMs offer more than a repurposing of old stuff
- Once we allow for imprecision, lots of doors open up
- For example, the gap between Bayes & non-Bayes is the feelings about prior dist'ns
 - Bayesians narrow down to a single prior
 - non-Bayesians can't narrow down at all
- An imprecise framework provides an opportunity to incorporate **some priors** exactly

Conclusion, cont.



- Especially important in high-dim problems
- Main-stream methods for regularization are awkward:
 - Fs' penalties aren't part of the model/assumptions
 - Bs' priors impose more structure than is justified
- New idea:
 - incorporates into the model exactly the structural assumptions that can be justified, no more & no less



ST 790 (001) Fall 2022 Advanced Special Topics

Imprecise-Probabilistic Foundations of Statistics & Data Science

[https://wordpress-courses2223.wolfware.ncsu.edu/
st-790-001-fall-2022/](https://wordpress-courses2223.wolfware.ncsu.edu/st-790-001-fall-2022/)

Questions?

- Quantities of interest often exist in the real world, not just as parameters in a posited statistical model
- e.g., quantiles, moments, optimal classifiers, ...
- One can posit a model and express the quantity of interest in terms of model parameters, but gives *indirect* inference:
 - maybe lots of nuisance parameters
 - potential issues with marginalization²⁹
 - risk of model misspecification bias
- *Direct* inference without going through a statistical model?

²⁹Fraser (Stat Sci 2011), M. arXiv:1607.05051

The idea that statistical problems do not have to be solved as one coherent whole is anathema to Bayesians but is liberating for frequentists. —Wasserman

- Non-Bayesians/frequentists have a lot of flexibility, often pretty easy to proceed without a model
- e.g., quantiles & moments estimated by sample counterparts
- Bayesians are constrained by their reliance on likelihood
- Here I'll describe a particular context in which generalizations of likelihood and Bayesian inference is possible

- Suppose we have iid data $Z^n = (Z_1, \dots, Z_n)$ from P
- Z_i 's could be scalars, vectors, (X, Y) pairs, etc.
- Loss function $(z, \theta) \mapsto \ell_\theta(z)$ measures compatibility of (z, θ)
- Risk function, $R(\theta) = \int \ell_\theta(z) P(dz)$, expected loss
- Quantity of interest: $\Theta = \arg \min_\theta R(\theta)$
- Example:
 - $\ell_\theta(z) = |z - \theta|$
 - $\arg \min R(\theta) = \text{median of } P$
- Empirical risk: $R_n(\theta) = n^{-1} \sum_{i=1}^n \ell_\theta(Z_i)$

Definition.

The *M-estimator* is a minimizer of the empirical risk

$$\hat{\theta}_n \in \arg \min_{\theta} R_n(\theta)$$

- Simple examples:
 - if $\ell_{\theta}(z) = |z - \theta|$, then $\hat{\theta}_n$ is the sample median
 - if $\ell_{\theta}(z) = (z - \theta)^2$, then $\hat{\theta}_n$ is the sample mean
- Generalizes maximum likelihood:
 - posit a model P_{θ} with PDF $p_{\theta}(z)$
 - define the log-loss $\ell_{\theta}(z) = -\log p_{\theta}(z)$
 - then the M-estimator is the MLE
 - applies when model is misspecified, so $\hat{\theta}_n$ is estimating the KL-minimizer $\arg \min_{\theta} K(P, P_{\theta})$

- Rough summary of M-estimator theory:
 - consistent under certain conditions
 - r_n -consistent under some more conditions
 - asymptotically normal under more still conditions
- This theory is pretty “standard,” lots of references
 - Huber
 - Casella & Berger
 - van der Vaart
 - Boos & Stefanski
 - ...
 - my book (eventually)
- I want to focus on something different...

Definition.

The *Gibbs posterior distribution* of Θ is

$$Q_n^{(\eta)}(d\theta) \propto \underbrace{e^{-\eta n R_n(\theta)}}_{\text{"likelihood"}} Q(d\theta)$$

where Q is a prior and $\eta > 0$ is a *learning rate* parameter

- Where does the learning rate $\eta > 0$ come from?
 - well-specified model $\rightarrow$ likelihood & prior are properly scaled
 - here, the scale of the loss is ambiguous, i.e., same problem if ℓ_θ is replaced by $k \times \ell_\theta$
 - must manually balance “likelihood” and prior $\implies \eta$
- Learning rate is practically important, *can't be ignored!*

- Data (X_i, Y_i) , $i = 1, \dots, n$, iid:³⁰
 - $X_i \in \mathbb{R}$ is a diagnostic measure on patient i ;
 - $Y_i \in \{\pm 1\}$, depending on whether patient i found the treatment to be effective/ineffective.
- Quantity of interest is the MCID:

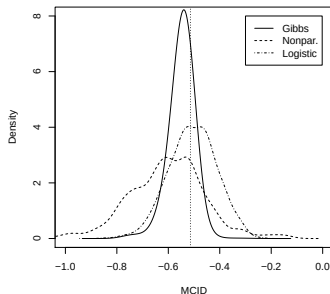
$$\Theta = \arg \min_{\theta} \underbrace{P\{Y \neq \text{sign}(X - \theta)\}}_{R(\theta)},$$

where P is the joint distribution of (X, Y) .

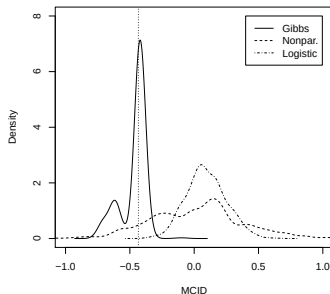
- MCID is the X -scale cutoff at which patients will tend to view the treatment as effective.
- Statistical versus practical/clinical significance...

³⁰Syring and M., (*JSPI* 2017, arXiv:1501.01840)

- Compare Gibbs ($\eta = 1$) to two Bayes posteriors
 - logistic regression
 - binary regression with nonparametric link



(e) "Good case"



(f) "Bad case"

- We expect that $Q_n^{(\eta)}$ will concentrate around Θ
- Intuition is clear: $Q_n^{(\eta)}$ follows R_n and R_n follows R
- Formal statement and proof requires more care³¹
- I'll focus here on finite-dim cases — it's easier, more intuitive, and allows for static choices of η
- Other different approaches:
 - Grünwald and Mehta (*JMLR* 2020)
 - Syring and M. (*Bernoulli* 2023, [arXiv:2012.04505](#))

³¹M. and Syring (*Handbook* 2022, [arXiv:2203.09381](#))

Theorem – Gibbs rates

Roughly, if there exists $\alpha > \beta > 0$ such that

$$\inf_{\theta: d(\theta, \Theta) > \delta} \{R(\theta) - R(\Theta)\} \gtrsim \delta^\alpha \quad \text{and} \quad \mathbb{E} \sup_{\theta: d(\theta, \Theta) < \delta} |\mathbb{G}_n(\ell_\theta - \ell_\Theta)| \lesssim \delta^\beta$$

and if the prior “puts enough mass near $\Theta = \Theta(P)$,” then

$$Q_n^{(\eta)}(\{\theta : d(\theta, \Theta) \gtrsim n^{-1/(2\alpha-2\beta)}\}) \rightarrow 0, \quad \text{in } P\text{-probability}$$

Theorem – Gibbs Bvm

Roughly, if the risk function is sufficiently smooth, etc., then

$$Q_n^{(\eta)} \approx N(\hat{\theta}_n, \{\eta n \ddot{R}(\hat{\theta}_n)\}^{-1}), \quad \text{for large } n$$

M. and Syring, (*Handbook* 2022, [arXiv:2203.09381](#))

- Cases with $n^{-1/2}$ rates and a BvM:
 - mean regression
 - (multivariate³²) quantiles
 - quantile regression
 - some classification problems
- MCID problem:
 - Key quantity: $f(x) = P(Y = 1 \mid X = x)$
 - Roughly, if f is γ -Holder near Θ , then rate $n^{-1/(1+2\gamma)}$
 - Specific cases:
 - if f is differentiable, then $\gamma = 1$ and rate is $n^{-1/3}$
 - if f has a jump at Θ , then $\gamma = 0$ and rate is n^{-1} .
- High- or infinite-dim cases too³³

³²Bhattacharya and M. (*JSPI* 2022, [arXiv:2002.01052](#))

³³Syring and M. (*Bernoulli* 2023, [arXiv:2012.04505](#))

- Asymptotic properties are nice, but there's a catch
- Unlike the *well-specified* Bayesian case, the above BvM does not imply good frequentist properties
- With a misspecified model, the Bayesian's results can be lousy³⁴ — and there's nothing he can do about it
- But Gibbsians have options, through tuning η
- Not perfect/one-size-fits-all, but better than nothing!

³⁴Kleijn and van der Vaart (EJS 2012)

- Recall the Gibbs posterior:

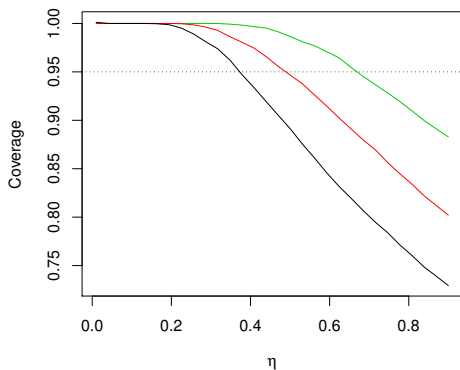
$$Q_n^{(\eta)}(d\theta) \propto e^{-\eta n R_n(\theta)} Q(d\theta).$$

- $Q_n^{(\eta)} \rightarrow \Theta$ for “all” η , so η must control the spread
- η too big makes credible sets too small and vice versa
- Tune η so that $Q_n^{(\eta)}$'s are credible regions are calibrated?
- e.g., 95% credible regions are 95% confidence regions?³⁵³⁶

³⁵Syring and M. (*Biometrika* 2019, [arXiv:1509.00922](#))

³⁶Wu and M. (*Bayesian Analysis* 2023, [arXiv:2012.11349](#))

MCID example: Coverage prob of 95% Gibbs posterior credible interval as a function of η for three n values



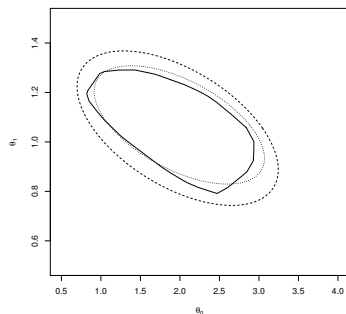
General Posterior Calibration (GPC) Algorithm:

- 0 Initialize η_0 and set $t = 0$
- 1 Sample new data sets from $\mathbb{P}_n$ to approximate the coverage prob of Gibbs credible region $C_{\eta_t, \alpha}$:
 - do Gibbs posterior computation for each new data set
 - check coverage using $\hat{\theta}_n = \arg \min R_n(\theta)$
- 2 If empirical coverage prob is close to $1 - \alpha$, then stop; otherwise, update $\eta_t \rightarrow \eta_{t+1}$ via stochastic approximation, increment t , and go back to Step 1.

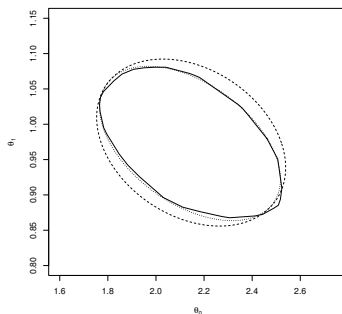
Syring and M. (*Biometrika* 2019, arXiv:1509.00922)

Credible region contours: Quantile regression with two-dim θ

- Gibbs + GPC (solid)
- bootstrapped M-estimator (dashed)



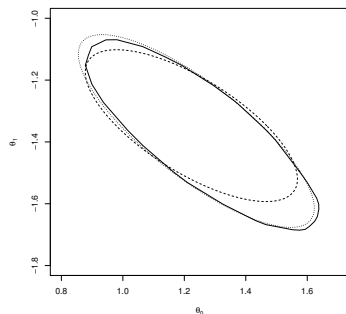
(g) $n = 50$, $\hat{\eta} \approx 0.96$



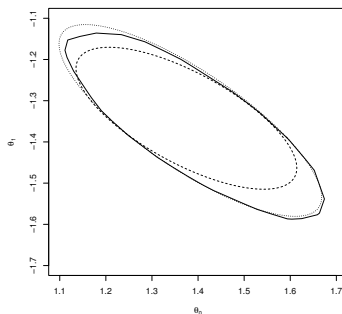
(h) $n = 400$, $\hat{\eta} \approx 0.90$

Credible region contours: Classification with two-dim θ

- Gibbs + GPC (solid)
- bootstrapped M-estimator (dashed)



(i) $n = 400$, $\hat{\eta} \approx 0.70$



(j) $n = 1000$, $\hat{\eta} \approx 0.58$

■ Two-dim quantile regression³⁷

BEL.s Bayesian empirical likelihood (Yang & He, *Annals* 2012)

Norm asymptotic normality

GPC Gibbs with GPC

■ GPC gives exact coverage and shorter length!

n		Coverage Prob $\times 100$			Avg Length $\times 100$		
		BEL.s	Norm	GPC	BEL.s	Norm	GPC
100	θ_0	97	95	95	106	100	91
	θ_1	98	98	95	58	55	47
400	θ_0	95	95	95	50	50	46
	θ_1	97	97	95	26	25	23
1600	θ_0	96	96	95	25	25	23
	θ_1	96	96	95	13	12	11

³⁷Syring and M. (*Biometrika* 2019, [arXiv:1509.00922](https://arxiv.org/abs/1509.00922))

- Gibbs is direct/liberated Bayes:
 - no misspecification bias
 - no nuisance parameters
 - no marginalization
- Good concentration rate properties
- Can tune η for valid uncertainty quantification
- Also possible to reverse engineer a suitable R :
 - image segmentation³⁸
 - Lévy density estimation³⁹
- More Gibbs stuff to do:
 - proof that the GPC algorithm “works”
 - low-dim structure learning
 - computation

³⁸Syring and M. (*Annals* 2020, [arXiv:1606.08400](#))

³⁹Wang and M., [arXiv:2109.06567](#)

Final remarks

- We covered a lot of material in this course!
- Not as deep as a full semester, but this is roughly the content of the first-year PhD course I teach
- Some of the material was “standard”
- But I hope that
 - at least some of the material was new to you
 - you learned something valuable
 - you enjoyed the experience

- The book is still in progress
- Aside from some to-be-filled-in exercises, I'll do chapters on
 - broader uncertainty quantification (IM stuff)
 - practical models (e.g., regression and GLMs)
 - non-model-based methods (e.g., Gibbs posteriors)
 - prediction problems
 - high-dim problems
- My goal is for this to be done by May 2025



ST 790 (001) Fall 2022 Advanced Special Topics

Imprecise-Probabilistic Foundations of Statistics & Data Science

[https://wordpress-courses2223.wolfware.ncsu.edu/
st-790-001-fall-2022/](https://wordpress-courses2223.wolfware.ncsu.edu/st-790-001-fall-2022/)

- Papers, talks, etc — www4.stat.ncsu.edu/~rmartin/
- Questions, etc — rgmarti3@ncsu.edu

Thanks for your attention!